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PREFACE TO THE SECOND EDITION

Since the first edition was published in 2000, *Managing Complications in Pregnancy and Childbirth* has been translated into several languages and today is used widely in training for and the provision of emergency obstetric care. The new edition brings the guidance in the manual into line with WHO's current recommendations for emergency obstetric and newborn care.

The updating process for the second edition involved WHO's Departments of Maternal, Newborn, Child and Adolescent Health and Reproductive Health and Research as well as MCSP, USAID's flagship maternal, newborn and child health program, led by Jhpiego. A core review group (see Acknowledgements) conducted a user survey to solicit feedback on use of the manual and suggestions to guide the revision. The core group then updated the following chapters with the current WHO recommendations:

- **Section 1: Clinical Principles:** Emotional and psychological support, Emergencies, General care principles, Antibiotic therapy, Operative care principles, Normal labour and childbirth, and Newborn care principles
- **Section 2: Symptoms:** Vaginal bleeding in early pregnancy; Vaginal bleeding after childbirth; Elevated blood pressure, headache, blurred vision, convulsions or loss of consciousness; Fever during pregnancy and labour; Fever after childbirth; Difficulty in breathing; Prelabour rupture of membranes; and Immediate newborn conditions or problems
- **Section 3: Procedures:** Induction and augmentation of labour, Manual removal of placenta, and Repair of vaginal and perineal tears

The updated chapters were reviewed at the first core group meeting in Washington, DC, in August 2015. The revised chapters were sent to an external panel of expert reviewers (see Acknowledgements). The external experts were selected on the basis of their clinical experience in emergency obstetric care in low- and middle-income countries. None of the external reviewers reported any conflict of interest.

The core group met again in Geneva in January 2016 to review comments from the external reviewers and to finalize the chapters. The finalized chapters were then sent to the WHO's Guidelines Review Committee for approval.

Minor revisions, including clarification of wording and corrections (e.g. grammatical and typographical), have been made throughout the manual. Some new figures were added to the manual and some figures have been slightly modified to make their meaning more clear.

It was beyond the scope of the second edition to address obstetric ultrasound. However, readers are referred to two recent WHO sources on obstetric ultrasound: Chapter 2 in the *Manual of Diagnostic Ultrasound*, Second Edition (WHO, 2013) and *WHO Recommendations on Antenatal Care for a Positive Pregnancy Experience* (WHO, 2016). The latter recommends one routine ultrasound scan before 24 weeks of gestation.

PREFACE TO THE FIRST EDITION

In support of the Safe Motherhood Initiative, the WHO Making Pregnancy Safer Strategy focuses on the Health Sector's contribution to reducing maternal and newborn deaths.

The Integrated Management of Pregnancy and Childbirth (IMPAC) is the technical component of the aforementioned strategy and mainly addresses the following:

- Improving the skills of health workers through locally adapted guidelines and standards for the management of pregnancy and childbirth at different levels of the health care system;
- Interventions to improve the health care system's response to the needs of pregnant women and their newborns, and to improve the district level management of health services, including the provision of adequate staffing, logistics, supplies and equipment;
- Health education and promotion of activities that improve family and community attitudes and practices in relation to pregnancy and childbirth.

This manual, and a similar one on the management of preterm and sick newborns, is written for midwives and doctors working in district hospitals. This manual complements and is consistent with the Essential Care Practice Guide for Pregnancy and Childbirth which is prepared mainly for the primary health care level. Together these manuals will provide guidance for health workers who are responsible for the care of pregnant women and newborns at all levels of care.

The interventions described in these manuals are based on the latest available scientific evidence. Given that evidence-based medicine is the standard on which to base clinical practice, it is planned to update the manual as new information is acquired.

It is hoped that this manual will be used at the side of the patient, and be readily available whenever a midwife or doctor is confronted with an obstetric emergency.

While most pregnancies and births are uneventful, all pregnancies are at risk. Around 15% of all pregnant women will develop a potentially life-threatening complication that calls for skilled care, and some will require a major obstetrical intervention to survive. This manual is written for midwives and doctors at the district hospital who are responsible for the care of women with complications of pregnancy, childbirth or the immediate postpartum period, including immediate problems of the newborn.

In addition to providing care to women in facilities, midwives and doctors have a unique role and relationship with:

- the community of health care providers within the district health system, including auxiliary and multipurpose health workers;
- family members of patients;
- community leaders;
- populations with special needs (e.g. adolescents, women with HIV/AIDS).

Midwives and doctors:

- support activities for the improvement of all district health services;
- strive for efficient and reliable referral systems;
- monitor the quality of health care services;
- advocate for community participation in health-related matters.

A district hospital is defined as a facility that is capable of providing quality services, including caesarean birth and blood transfusion. Although many of the procedures in this manual require specialized equipment and the expertise of specially trained providers, it should be noted that many of the life-saving procedures described can also be performed at health centres.

A woman presenting with a life-threatening obstetric complication is in an emergency situation requiring immediate diagnosis and management. Therefore, the main text of this manual is arranged by **symptom** (e.g. vaginal bleeding in early pregnancy). Because this symptom-based approach is different than most medical texts, which are arranged by disease, an indexed list of diagnoses is provided.

The emphasis of the manual is on rapid assessment and decision making. The clinical action steps are based on clinical assessment, with limited reliance on laboratory or other tests that would be unavailable in many field situations, and most are possible in a variety of clinical settings (e.g. district hospital or health centre).

Section 1 outlines the **clinical principles** of managing complications in pregnancy and childbirth. It begins with a table that the health care worker can use to rapidly assess the woman's condition and initiate appropriate treatment. This section includes the general principles of emergency, general and operative care, including infection prevention, the use of blood and replacement fluids, antibiotics and anaesthesia and analgesia. In order to provide health care workers with the information needed to differentiate between normal processes and a complication, the section also includes a description of normal labour and childbirth, including use of the partograph and active management of the third stage. Guidance on the initial care of the normal newborn is also provided. Finally, this section also outlines the linkage between providers and their community and how to provide emotional support to the woman and her family.

Section 2 describes the **symptoms** by which women with complications of pregnancy and childbirth present. The symptoms reflect the major causes of mortality and morbidity. For each symptom there is a statement of general, initial management. Where relevant, diagnosis tables lead to identifying the diagnosis which is causing the symptom. Simplified management protocols for these specific diagnoses then follow. Where there are several choices of therapy, the most effective and inexpensive is chosen. Also in this section is information on management for immediate (within the first 24 hours) conditions or problems of the newborn.

Section 3 describes the **procedures** that may be necessary in the management of complications of pregnancy and childbirth. These procedures are not intended to be detailed “how-to” instructions but rather a summary of the main steps associated with each procedure. Because general operative care principles are summarized in Section 1, these are not repeated for each procedure, unless there is care required that is specific to the procedure (e.g. post-procedure care for ketamine anaesthesia). Clear guidance is provided on drugs and dosages, a wide variety of anaesthesia

options (e.g. safe caesarean under local anaesthesia) and safe, effective and lower-cost techniques (e.g. single layer closure of the uterus).

A list of **essential drugs** is included as an appendix to the manual. Finally, the manual's **index** is organized so that it can be used in an emergency situation to find relevant material quickly. The most critical information, including diagnosis, management and steps for a procedure, is listed first in bold. Other relevant entries follow in alphabetical order. Only the pages containing critical or relevant information are included, rather than every page that contains a word or phrase.

ABBREVIATIONS

xv

ACT	artemisinin-based combination therapy
AIDS	acquired immunodeficiency syndrome
BP	blood pressure
BCG	bacillus Calmette-Guérin
CPAP	continuous positive airway pressure
D and E	dilatation and evacuation
HELLP	haemolysis, elevated liver enzymes and low platelets
HIV	human immunodeficiency virus
IM	intramuscular
IPTp	intermittent preventive treatment in pregnancy
IUD	intrauterine device
IV	intravenous
NASG	non-pneumatic anti-shock garment
PPH	postpartum haemorrhage
PPROM	preterm prelabour rupture of membranes
PROM	prelabour rupture of membranes
dL	decilitre
g	gram
kg	kilogram
L	litre
mcg	microgram
mg	milligram
mIU	milli-international units
mL	millilitre

LIST OF DIAGNOSES

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Maternal/Fetal

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Abruptio placentae	S-23	Haemorrhage, postpartum	S-29
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Amnionitis	S-163	Heart failure	S-151
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Anaphylaxis	C-52	Hepatitis	S-115
Appendicitis	S-139	Hypoglycaemia	S-125
Atelectasis	S-129	Inadequate uterine activity	S-83
Atonic uterus	S-32	Incomplete abortion	S-12
Breast abscess	S-133	Inevitable abortion	S-12
Breast engorgement	S-132	Inverted uterus	S-45
Breast infection	S-133	Large fetus	S-102
Breech presentation	S-95	Malaria, severe	S-121
Bronchial asthma	S-150	Malaria, uncomplicated	S-118
Brow presentation	S-89	Malpresentation or malposition	S-85
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Coagulopathy	S-24	Multiple pregnancy	S-102
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Compound presentation	S-94	Normal labour	C-77
Cystitis	S-116	Obstructed labour	S-83
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Pre-eclampsia		Asphyxia, newborn	S-166
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Prolapsed cord	S-111	breathing difficulty	S-174
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Prolonged expulsive phase	S-84	Low birth weight or	
Prolonged latent phase	S-81	moderately preterm baby	S-176
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Pulmonary oedema	S-150	transmission of syphilis	S-183
Retained placenta or		Newborn sepsis	S-180
placental fragments	S-43	Very low birth weight or	
Ruptured uterus	S-24	very preterm baby	S-177
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Shoulder presentation	S-97		
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perineum	S-43		
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Vaginitis	S-161		
Wound abscess, wound			
seroma or wound			
haematoma	S-135		
Wound cellulitis	S-135		

RAPID INITIAL ASSESSMENT

C-1

The survival of a woman experiencing an obstetric emergency is determined by the amount of time it takes for care to be delivered and by the level and quality of care provided. When a woman of childbearing age presents with a problem, rapidly assess her condition to determine the extent of her illness.

TABLE C-1. Rapid initial assessment^a

Assess	Danger Signs	Consider
Airway and breathing	LOOK FOR: • cyanosis (blueness) • respiratory distress EXAMINE: • skin: pallor • lungs: wheezing or rales	• severe anaemia • heart failure • pneumonia • asthma See Difficulty in breathing, page S-149
Circulation (signs of shock)	EXAMINE: • skin: cool and clammy • pulse: fast (110 or more) and weak • blood pressure: low (systolic less than 90 mmHg)	Shock, page S-1
Vaginal bleeding (early or late pregnancy or after childbirth)	ASK IF: • pregnant, length of gestation • recently given birth • placenta delivered EXAMINE: • vulva: amount of bleeding, placenta retained, obvious tears • uterus: atony • bladder: full Do not do a vaginal exam at this stage.	• abortion • ectopic pregnancy • molar pregnancy See Vaginal bleeding in early pregnancy, page S-7 • abruptio placentae • ruptured uterus • placenta praevia See Vaginal bleeding in later pregnancy and labour, page S-21 • atonic uterus • tears of cervix and vagina • retained placenta • inverted uterus See Vaginal bleeding after childbirth, page S-29

Assess	Danger Signs	Consider
Unconscious or convulsing	ASK IF: • pregnant, length of gestation EXAMINE: • blood pressure: high (diastolic 90 mmHg or more) • temperature: 38°C or more	• eclampsia • malaria • epilepsy • tetanus See Elevated blood pressure, heading, blurred vision, convulsions or loss of consciousness, page S-49
Dangerous fever	ASK IF: • weak, lethargic • frequent, painful urination EXAMINE: • temperature: 38°C or more • unconscious • neck: stiffness • lungs: shallow breathing, consolidation • abdomen: severe tenderness • vulva: purulent discharge • breasts: tender	• urinary tract infection • malaria See Fever during pregnancy and labour, page S-113 • endometritis • pelvic abscess • peritonitis • breast infection See Fever after childbirth, page S-127 • complications of abortion See Vaginal bleeding in early pregnancy, page S-7 • pneumonia See Difficulty in breathing, page S-149
Abdominal pain	ASK IF: • pregnant, length of gestation EXAMINE: • blood pressure: low (systolic less than 90 mmHg) • pulse: fast (110 or more) • temperature: 38°C or more • uterus: state of pregnancy	• ovarian cyst • appendicitis • ectopic pregnancy See Abdominal pain in early pregnancy, page S-137 • possible term or preterm labour • amnionitis • abruptio placentae • ruptured uterus See Abdominal pain in later pregnancy and after childbirth, page S-141

^a This list does not include all of the problems a woman might face in pregnancy or the puerperal period. It identifies those problems that put the woman at greatest risk of maternal morbidity and mortality.

The woman also needs **prompt attention** if she has any of the following signs:

- blood-stained mucus discharge (show) with palpable contractions
- ruptured membranes
- pallor
- weakness
- fainting
- severe headaches
- blurred vision
- vomiting
- fever
- respiratory distress.

Send the woman to the front of the queue and treat promptly.

IMPLEMENTING A RAPID INITIAL ASSESSMENT SCHEME

Delayed activation of a rapid response to an obstetric emergency is associated with an increased risk of death. Rapid initiation of treatment requires immediate recognition of the specific problem and quick action. This can be done by:

- clearly identifying women presenting for care who warrant prompt or immediate attention from a health worker, including those waiting for routine consultations who should pass to the front of the queue;
- having norms and protocols (and knowing how to use them) to outline how to recognize a genuine emergency and react immediately, including the roles and responsibilities of all members of staff in the facility;
- training all staff—including clerks, guards, door keepers and switchboard operators—to react in an agreed upon fashion (e.g. “sound the alarm,” call for help) when a woman arrives at the facility with an obstetric emergency or pregnancy complication or when the facility is notified that a woman is being referred;

- ensuring that an emergency trolley with equipment, medications and supplies is accessible (keys are available) and ready to use at all times (see **page C-21**), that equipment is in working order (daily checks) and that staff are properly trained to use it;
- conducting clinical or emergency drills with staff to ensure their readiness at all levels;
- debriefing team members after an event in an effort to evaluate and improve their response; and
- agreeing on schemes by which women with emergencies can be exempted from payment, at least temporarily (e.g. local insurance schemes, health committee emergency funds).

Pregnancy is typically a time of joy and anticipation. It can also be a time of anxiety and concern. Listening and talking respectfully and sensitively with a woman and her family can help build the woman's trust and confidence in her health care providers.

Women who develop complications may have difficulty talking to the provider and explaining their problem. It is the responsibility of the entire health care team to speak with the woman respectfully and put her at ease. Focusing on the woman means that the health care provider and staff:

- respect the woman's dignity and right to privacy;
- respect the woman's right to information and informed consent;
- respect the woman's right to decline any treatment or procedures offered;
- respect the woman's choices and preferences, including companionship during maternity care, procedures and treatment;
- protect a woman's privacy rights and protections with respect to her health information, including how her health information is used and to whom her information is disclosed by health care providers;
- are sensitive and responsive to the woman's needs;
- are nonjudgmental about the decisions that the woman and her family make regarding the woman's care.

It is understandable to disagree with a woman's risky behaviour or a decision that has resulted in a delay in seeking care. It is not acceptable, however, to show disrespect for a woman or disregard for a medical condition that you believe is a result of her behaviour. Provide respectful corrective counselling after the complication has been dealt with, not before or during management of the problem.

RIGHTS OF WOMEN

Providers should be aware of the rights of women when receiving maternity care services:

- Every woman receiving care has the right to information about her health.
- Every woman has the right to discuss her concerns in an environment in which she feels confident.
- A woman should know in advance the type of procedure that is going to be performed.

- A woman (or her family, if necessary) should give informed consent before the provider performs any procedure.
- A woman (or her family, if necessary) has the right to decline any treatment or procedure offered.
- Procedures should be conducted in an environment (e.g. labour ward) in which the woman's right to privacy is respected.
- A woman has the right to determine how her health information is used and to whom her information is disclosed by health care providers.
- A woman should be made to feel as comfortable as possible when receiving care.
- A woman has the right to express her views about the services she receives.

When a provider talks to a woman about her pregnancy or a complication, s/he should use basic communication techniques. These techniques help the provider establish an honest, caring and trusting relationship with the woman. If a woman trusts her provider and feels that the provider has her best interests at heart, she will be more likely to return to the facility for childbirth or come early if there is a complication.

COMMUNICATION TECHNIQUES

Speak in a calm, quiet manner and assure the woman that the conversation is confidential. Be sensitive to any cultural or religious considerations and respect her views. In addition:

- Ask the woman whom she would like present. Facilitate the presence of only those she chooses to be present.
- Encourage the woman and her family to speak honestly and completely about events surrounding the complication.
- Listen to what the woman and her family have to say and encourage them to express their concerns; try not to interrupt.
- Respect the woman's sense of privacy and modesty by closing the door or drawing curtains around the examination table.
- Let the woman know that she is being listened to and understood.
- Use supportive nonverbal communication such as nodding and smiling.

- Answer the woman's questions directly and in a calm, reassuring manner.
- Explain what steps will be taken to manage the situation or complication.
- Ask the woman to repeat back to you the key points to ensure her understanding.

If a **woman must undergo a surgical procedure**, explain to her the nature of the procedure and its risks and help to reduce her anxiety. Women who are extremely anxious have a more difficult time during surgery and recovery.

For more information on providing emotional support during an emergency, see **page C-9**.

EMOTIONAL AND PSYCHOLOGICAL SUPPORT IN OBSTETRIC AND NEWBORN EMERGENCIES

C-9

In every country and community in the world, pregnancy and childbirth are momentous events in the lives of women and families, and represent a time of intense vulnerability (White Ribbon Alliance, *Respectful Maternity Care: The Universal Rights of Childbearing Women*, 2012). The concept of safe motherhood is usually restricted to physical safety, but childbearing is also an important rite of passage, which may have deep personal and cultural significance for a woman and her family. The notion of safe motherhood must be expanded beyond the prevention of morbidity and mortality to encompass respect for women's basic human rights, including women's autonomy, dignity, feelings, and choices and preferences, including the choice of companionship, wherever possible.

GENERAL PRINCIPLES OF COMMUNICATION AND SUPPORT

Emergency situations are disturbing for all concerned, and they evoke a range of emotions that can have significant consequences. Because each emergency is unique, this chapter offers general guidance for responding to emotional and psychosocial needs in emergencies. Honest communication and genuine empathy are probably the most important keys to effective care in such situations. In addition, health care staff must try to understand and acknowledge the specific needs and cultural practices of individual women and their families, including the potentially differing needs of individual family members.

In general, women need opportunities to talk about their pregnancy and birth experience(s) and to be reassured that they will be well cared for during pregnancy, childbirth and the postpartum period. Good communication encompasses the following: an explanation of events; an explanation of findings and exploration of courses of action that will assist the woman in making informed decisions; an opportunity to discuss feelings such as loss, fear, anger and self-blame; social support; and a discussion of possible future childbearing.

EMOTIONAL AND PSYCHOLOGICAL REACTIONS

Multiple factors may influence women's psychological status before, during and after pregnancy. A woman's psychological status in turn influences her ability to care for herself and her baby and to follow health care recommendations. Therefore, to achieve optimal outcomes, it is as

important to provide emotional and psychological support as it is to provide medical care.

Common reactions to an obstetric/newborn emergency or death include:

- denial (feelings that “it can’t be true”);
- guilt regarding possible responsibility;
- anger (frequently directed toward health care staff but often masking anger that parents direct at themselves for “failure”);
- blame, fear, anxiety, sorrow, grief and a sense of failure;
- bargaining (particularly if the woman/baby hovers for a while between life and death);
- depression and loss of self-esteem, which may be long-lasting;
- isolation (feelings of being different or separate from others), which may be reinforced by caregivers who might avoid people who experience loss; and disorientation.

How each member of the family reacts to an emergency situation depends on:

- the woman’s support systems, including her relationship with her partner and the presence of family or supportive individuals;
- the social situation of the woman/couple and their cultural and religious practices, beliefs and expectations;
- the woman’s previous pregnancy outcomes, chronic medical problems and history of psychiatric problems, sexual and domestic abuse, or substance abuse;
- the personalities of the people involved and the quality and nature of social, practical and emotional support; and
- the nature, gravity and prognosis of the problem and the availability and quality of health care services.

Awareness and recognition of these factors can help maternal health care providers better support women and their families. Failure to recognize risk factors and/or psychological distress in pregnant and postpartum women can have devastating and tragic consequences for the woman and her child(ren) and family.

- In cases of severe deformity, wrap the baby before giving her/him to the mother to hold so that she can see the normality of the baby first. Do not force the mother to examine the abnormality.
- Provide a bed or cot in the room so the companion can stay with the woman if she chooses.

AFTER THE EVENT

- Discuss the baby and the problem with the woman and her family together, if possible.
- Ensure that the woman and her partner have free access to their baby. Keep the baby with the mother at all times. The more the woman and her partner can do for the baby themselves, the more quickly they will accept the baby.
- Where relevant (e.g. abnormalities associated with drug use, alcohol consumption or smoking during pregnancy; advanced maternal age; untreated viral or bacterial infections), arrange a discussion with both the woman and her partner about possible preventive measures for the future, without blaming the woman/family.
- Ensure access to supportive professional individuals and groups.
- Provide supportive services for health care providers as a way to relieve anxiety and distress.

PSYCHOLOGICAL MORBIDITY

Postpartum emotional distress is fairly common after pregnancy and ranges from mild postpartum blues (affecting about 80% of women) to postpartum depression or psychosis. Postpartum psychosis can pose a threat to the life of the woman and the baby.

“Postpartum blues” refers to mild depressive symptoms (i.e. sadness, tearfulness, irritability and anxiety), insomnia and decreased concentration. The symptoms of postpartum blues develop within two to three days of giving birth and typically peak over the next few days and resolve within two weeks. Women with postpartum blues are at increased risk of developing postpartum minor depression or major depression.

POSTPARTUM DEPRESSION

Postpartum depression affects up to 34% of women. It typically occurs in the early postpartum weeks or months and may persist for a year or more. Depression is not necessarily one of the leading symptoms, although it is usually evident. Other symptoms include exhaustion, irritability, weepiness, low energy and motivational levels, feelings of helplessness and hopelessness, loss of libido, and appetite and sleep disturbances. Headache, asthma, backache, vaginal discharge and abdominal pain may be reported. Symptoms might also include obsessional thinking, fear of harming the baby or self, suicidal thoughts, and depersonalization.

The prognosis for postpartum depression is good with early diagnosis and treatment. More than two thirds of women recover within a year. Facilitating the presence of a companion of choice during pregnancy, labour, childbirth and the postpartum period may help prevent postpartum depression.

Women should routinely be screened for depression during postpartum visits:

- At each postnatal contact, women should be asked about their emotional well-being, what family and social support they have, and their usual coping strategies for dealing with day-to-day matters. All women and their families/partners should be encouraged to tell their health care professional about any changes in mood, emotional state or behaviour that are outside the woman's normal pattern.
- At 10–14 days after birth, all women should be asked about resolution of mild, transitory postpartum depression ("maternal blues"/"baby blues"). **If symptoms have not resolved**, the woman's psychological well-being should continue to be assessed for postnatal depression and, if symptoms persist, further evaluated.

Women who screen positive for postnatal depression should be linked to ongoing care, including further diagnostic evaluation and ongoing treatment as indicated. In addition, surveillance for maternal depression can be performed at pediatric visits in the first 12 months postpartum.

Risk factors for postpartum depression include:

- Previous postpartum depression;
- Active or previous mental illness;
- Being a member of a vulnerable population;
- Traumatic childbirth;

- Giving birth to a preterm or stillborn infant or experiencing the death of a newborn;
- Having an infant admitted to intensive care;
- History of being a neglected child.

Once established, postpartum depression requires psychological counselling and ongoing practical support. In general:

- Psychosocial interventions should be the first line of management of depression during pregnancy and breastfeeding. Antidepressant medication should be avoided as much as possible in this population.
- Listen to the woman and provide encouragement.
- Incorporate general principles of care: provide empathic, clear and sensitive communication with women and their families, and mobilize and provide social support.
- Assure the woman that the experience is fairly common; many other women experience the same thing and most women are able to recover.
- Provide psycho-education and avoid over-medicalization. To avoid stigma, use commonly understood terms such as *stress* and *burden* instead of *depressive disorder* or *illness*.
- Assist the woman in rethinking her image of motherhood, and assist the couple in thinking through their respective roles as new parents. They may need to adjust their expectations and activities.
- Provide adjunct treatments such as structured physical activity, relaxation and problem-solving, if available, and integrate these activities into the woman's daily routine.
- Facilitate social support for the woman throughout the intervention.
- Give attention to the woman's overall well-being, attending to her physical as well as psychological health and emphasizing the importance of the mother-infant relationship in this period.
- Provide practical help (with the baby and with home care).
- **If depression is severe**, women may benefit from treatment with an antidepressant medication. Be aware that medication can be passed through breastmilk and that decisions about treatment will need to take into consideration and address a woman's breastfeeding status. Where possible, consult a specialist.

Care can be home-based or can be offered through day care clinics, with links established for community-based care and other mental health services. Local support groups of women who have had similar experiences can be valuable for many women.

POSTPARTUM PSYCHOSIS

Postpartum psychosis typically occurs around the time of childbirth, most often within the first two weeks postpartum, and affects less than 1% of women. Severe postpartum depression may be associated with psychosis. Symptoms of postpartum psychosis include delusions, hallucinations, sleep disturbances, obsessive thoughts about the baby, severe depression, anxiety, despair, and suicidal or infanticidal impulses. Women who have bipolar disorder or schizoaffective disorder have a higher risk of developing postpartum psychosis. About half of women experiencing postpartum psychosis have a history of mental illness.

The prognosis for recovery is excellent, but about 50% of women will suffer a relapse with subsequent births. Postpartum psychosis is a mental health emergency and requires immediate attention to prevent the woman from harming herself or her baby. In general:

- Immediately seek comprehensive psychiatric and medical care.
- A woman with active psychosis should not care for her infant “as usual.” Arrangements should be made for supplemental care to ensure the safety of the newborn.
- Family members should remain with the woman at all times.
- Often, hospitalization is required for women with postpartum psychosis.
- Provide psychological support and arrange for practical help with the baby and with home care, including measures to ensure that the baby remains safe until the psychosis has resolved.
- Listen to the woman and provide support for her and her family, including linkages for further diagnostic evaluation and treatment, as indicated. This is important for avoiding tragic outcomes.
- Lessen stress.
- Avoid dealing with emotional issues when the woman is unstable.
- If **antipsychotic medications are used**, be aware that medication can be passed through breastmilk, so decisions about treatment must take into consideration and address a woman’s breastfeeding status.

Emergencies can happen suddenly, as with a convulsion, or they can develop as a result of a complication that was not detected or is not properly managed or monitored.

PREVENTING EMERGENCIES

Most emergencies can be prevented by:

- careful planning;
- empowering women and communities;
- following clinical guidelines;
- closely monitoring the woman; and
- responding immediately when problems are identified.

PREPARING FOR EMERGENCIES

To improve first response in case of an obstetric emergency, facilities should ensure the availability of:

- essential equipment, medications and supplies;
- skilled staff;
- a system for responding to emergencies; and
- regular drills.

An emergency trolley should be available and ready to use at all times. Providers must be prepared to respond quickly at all times to the most common obstetric and newborn emergencies. The emergency trolley should be checked daily and between shifts to make sure that the equipment is functional, that drugs have not expired and that everything that should be on the trolley is actually there. The person in charge of the unit should set up a schedule that is clearly posted and that indicates who is responsible for checking the trolley. The trolley should be restocked and any equipment should be appropriately treated immediately after responding to every emergency.

A facility's ability to manage emergencies should be assessed and reinforced by frequent emergency drills. A schedule should be set up so that units regularly practice the drill and become proficient at responding to emergencies. It is important to know what to do—and how to do it—in any particular emergency. It is also important to communicate when

responding to the emergency, to work as a team, and to carefully document what happened and what medical care was provided.

Everyone responding to an emergency should learn to talk out loud to each other while they are working, so that each provider knows what the other is doing and what they have found. Talking out loud can help coordinate efforts in response to the emergency, and can help avoid duplication of effort.

RESPONDING TO AN EMERGENCY

Responding to an emergency promptly and effectively requires that members of the clinical team know their roles and how the team should function. Team members should also know:

- clinical situations and their diagnoses and treatments;
- medicines, how they are used and administered and their side effects; and
- emergency equipment and how it functions.

The ability of a facility to deal with emergencies should be assessed and reinforced by frequent practice emergency drills.

When managing an emergency:

- Introduce yourself.
- Ask the woman her name; if she is unconscious, ask for the woman's name from her companion.
- Encourage the companion to stay with the woman.
- Explain all procedures; ask permission and keep the woman informed as much as you can about what you are doing. If the woman is unconscious, talk to her companion.
- Ensure and respect the woman's privacy during examination and discussion.
- Do not leave the woman unattended.
- Ensure that the emergency transport the woman used to get to the health facility is retained at the facility until a clear plan of management is in place.

INITIAL MANAGEMENT OF AN OBSTETRIC EMERGENCY

When a woman presents with an obstetric emergency:

- Stay calm. Think logically and focus on the needs of the woman.
- Take charge. Avoid confusion by choosing one person to be in charge.
- **SHOUT FOR HELP.** Have one person go for help and have another person gather emergency equipment and supplies (e.g. oxygen cylinder, emergency kit).
- If the **woman is unconscious**, assess the airway, breathing and circulation. Begin resuscitation of the woman, as needed (e.g. assist breathing, start intravenous infusion).
- If **shock is suspected**, immediately begin treatment (**page S-1**). Even if signs of shock are not present, keep shock in mind as you evaluate the woman further, because her status may worsen rapidly. If **shock develops**, it is important to begin treatment immediately.
- Position the woman lying down on her left side with her feet elevated. Loosen tight clothing.
- Talk to the woman and help her stay calm. Ask her or someone with her what happened, what symptoms she is experiencing and when they started.
- Perform a **rapid evaluation** of the woman's general condition, level of consciousness, presence of anxiety and/or confusion, blood loss, colour, and temperature of skin (**page C-1**).
- Put in one or two large-bore IV cannulae/needles.
- Get blood samples before infusing IV fluids.
- Infuse IV fluids. Adjust the flow rate based on the woman's condition and potential risks of fluid overload (e.g. pre-eclampsia).
- Perform a rapid targeted history and physical examination to make a differential diagnosis of the problem.
- Stabilize the woman.
- Consider catheterization if monitoring urinary output or a distended bladder.
- Document history, findings, actions and plan for continued management based on the cause of the emergency.

BASIC PRINCIPLES WHEN PROVIDING CARE

- Where feasible, ensure that the woman has a companion of her choice with her.
- Provide information to the woman—and any accompanying family members the woman would like to be involved in decision-making—about diagnostic tests to be performed, supportive care to be provided (e.g. IV infusion), the process of care, her diagnosis, treatment options and the estimated time for in-patient care if required.
- If the woman is unconscious, explain the procedure to her family.
- Obtain informed consent for any procedures, diagnostic or therapeutic, and care.

INFECTION PREVENTION AND CONTROL

- In the context of maternal and newborn health care, infection prevention and control has three primary objectives:
 - to prevent major infections when providing services;
 - to minimize the risk of transmitting serious diseases such as hepatitis B and HIV/AIDS to the women and to health care providers and staff, including cleaning and housekeeping personnel; and
 - to protect the environment by properly disposing of medical waste.
- The recommended infection prevention and control practices are based on the following principles:
 - Consider every person (woman or staff) as potentially infectious.
 - Handwashing is the most practical procedure for preventing cross-contamination.
 - Wear gloves before touching anything wet—broken skin, mucous membranes, blood or other body fluids (secretions or excretions).
 - Use barriers (e.g. protective goggles, face masks or aprons) if splashes and spills of any body fluids (secretions or excretions) are anticipated.
 - Use safe work practices such as not recapping or bending needles, proper instrument processing, and proper disposal of medical waste.

HANDWASHING

- Handwashing with water and soap removes visible dirt and organic material as well as bacterial contamination.
- Thoroughly rub together all surfaces of the hands lathered with plain or antimicrobial soap. Wash for 15–30 seconds and rinse with a stream of running or poured water.
- Wash hands:
 - before and after examining the woman (or having any direct contact);
 - after exposure to blood or any body fluids (secretions or excretions), even if gloves were worn;
 - before putting gloves on because hands may contaminate gloves;
 - after removing gloves because the gloves may have holes in them.
- To encourage handwashing, programme managers should make every effort to provide soap and a continuous supply of clean, potable water, from either the tap or a bucket, and single-use towels. Do not use shared towels to dry hands.
- To wash hands for surgical procedures, see **page C-67**.

FIGURE C-1. Hand hygiene technique with soap and water

Duration of the entire procedure: 40-60 seconds

0		Wet hands with water;
1		Apply enough soap to cover all hand surfaces;
2		Rub hands palm to palm;
3		Right palm over left dorsum with interlaced fingers and vice versa;
4		Palm to palm with fingers interlaced;
5		Backs of fingers to opposing palms with fingers interlocked;
6		Rotational rubbing of left thumb clasped in right palm and vice versa;

7		Rotational rubbing, backwards and forwards with clasped fingers of right hand in left palm and vice versa;
8		Rinse hands with water;
9		Dry hands thoroughly with a single use towel;
10		Use towel to turn off faucet;
11		You hands are now safe

HANDRUB WITH ALCOHOL-BASED SOLUTION

- If hands are not visibly dirty, use handrub with an alcohol-based solution for antimicrobial activity.
- Use plenty of handrub, enough to cover both hands, and rub all parts of the hands for 30 seconds or until hands are dry.

Wash hands with soap and water when visibly soiled.

Note: Remove rings, wrist watches and bracelets. Avoid wearing artificial nails because they increase the risk of hands remaining contaminated even after washing with soap and water and using alcohol-based handrub.

FIGURE C-2. Hand hygiene technique with alcohol-based handrub

Duration of the entire procedure: 20-30 seconds

1a		Apply a palmful of the product in a cupped hand, covering all surfaces;
1b		
2		Rub hands palm to palm
3		Right palm over left dorsum with interlaced fingers and vice versa;
4		Palm to palm with fingers interlaced;
5		Backs of fingers to opposing palms with fingers interlocked;
6		Rotational rubbing of left thumb clasped in right palm and vice versa;

7		Rotational rubbing, backwards and forwards with clasped fingers of right hand in left palm and vice versa;
8		Once dry, your hands are safe.

GLOVES AND GOWNS

- The use of gloves does not replace the need for hand hygiene by either handrubbing or handwashing.
- Wear gloves:
 - when there is a chance of touching blood, body fluids, secretions or excretions;
 - when performing a procedure (**Table C-2, page C-31**);
 - when handling soiled instruments, gloves or other items soiled with body fluids; and
 - when disposing of contaminated waste items (cotton, gauze or dressings).
- Wash hands before retrieving gloves from a box to prevent introducing skin commensals and pathogenic bacteria into glove boxes.
- When wearing gloves, change or remove them when moving from a contaminated body site to another body site (including a mucous membrane, non-intact skin or medical device within the same woman or the environment).
- Change gloves between caring for different individuals. A separate pair of gloves must be used for each woman to avoid cross-contamination.
- Immediately remove gloves after completing a procedure or caring for a woman. Do not walk around with contaminated gloves as this will contaminate the environment.

- Remove gloves by turning the first glove inside out as it is pulled over the hand. During the removal of the second glove, avoid touching the outer surface by slipping the fingers of the ungloved hand under the glove and pulling it inside out as it is pulled over the hand.
- Immediately dispose of the used gloves in a lined waste container. The reuse of gloves is not recommended. Disposable gloves are preferred.
- Immediately clean hands with handrub or soap and water after removing gloves.
- A clean, but not necessarily sterile, gown should be worn during all childbirth procedures:
 - If the gown has long sleeves, the gloves should be put on over the gown sleeve to avoid contamination of the gloves.
 - Ensure that gloved hands (sterile) are held above the level of the waist and do not come into contact with the gown.
- In addition to the gown, goggles and mask should be worn as a measure of protection.

TABLE C-2. Glove and gown requirements for common obstetric procedures^a

Procedure	Preferred Gloves	Alternative Gloves ^b	Gown
Blood drawing, starting IV infusion	Exam ^c	Sterile surgical ^d	None
Pelvic examination	Exam	Sterile surgical	None
Manual vacuum aspiration, dilatation and curettage, colpotomy, culdocentesis	Sterile surgical	None	None
Laparotomy and intra-abdominal procedures, artificial rupture of membranes, normal childbirth, instrumental birth, episiotomy, repair of cervical or perineal tears, craniotomy, craniocentesis, bimanual compression of uterus, manual removal of placenta, correcting uterine inversion	Sterile surgical	None	Clean, high-level disinfected or sterile

Procedure	Preferred Gloves	Alternative Gloves ^b	Gown
Handling and cleaning instruments	Utility ^c	Exam	None
Handling contaminated waste	Utility	Exam	None
Cleaning blood or body fluid spills	Utility	Exam	None

^a Gloves and gowns are not required to be worn to check blood pressure or temperature, or to give injections.

^b Alternative gloves are generally more expensive and require more preparation than preferred gloves.

^c Exam gloves are single-use, disposable latex gloves.

^d Surgical gloves are latex gloves that are sized to fit the hand.

^e Utility gloves are thick household gloves.

HANDLING SHARP INSTRUMENTS AND NEEDLES

OPERATING THEATRE AND LABOUR WARD

- Do not leave sharp instruments or needles (“sharps”) in places other than “safe zones” (page C-70).
- Alert other workers before passing sharps.

HYPODERMIC NEEDLES AND SYRINGES

- Use each needle and syringe only once.
- Promptly dispose of needles and syringes in a puncture-proof container.
- Do not disassemble a needle and syringe after use.
- Do not recap, bend or break needles prior to disposal.
- Make hypodermic needles unusable by burning them.

WASTE DISPOSAL

- The purpose of waste disposal is to:
 - minimize the spread of infections and reduce the risk of accidental injury to staff, clients and visitors;
 - protect those who handle waste from infection and accidental injury;
 - prevent the spread of infection to the local community; and
 - reduce the likelihood of contamination of the soil or ground water with chemicals or microorganisms.
- Health facilities generate four kinds of waste: noncontaminated waste, sharps, nonsharps contaminated waste and hazardous waste. Waste should be sorted, placed into separate waste containers and disposed of appropriately:
 - Noncontaminated waste (e.g. paper from offices, boxes) poses no infectious risk and can be disposed of according to local guidelines.
 - Proper handling of contaminated waste (blood- or body fluid–contaminated items) is required to minimize the spread of infection to hospital personnel and the community. Proper handling means:
 - wearing utility gloves;
 - transporting solid contaminated waste to the disposal site in covered containers;
 - disposing of all sharp items in puncture-proof containers;
 - carefully pouring liquid waste down a drain or flushable toilet (dedicated toilet that is not one used by clients or staff);
 - burning or burying contaminated solid waste; and
 - washing hands, utility gloves and containers after disposal of infectious waste.

STARTING AN IV INFUSION

- Explain what is involved in giving an IV infusion and its purpose to the woman and any accompanying family members the woman would like to have involved in decision-making. If the woman is unconscious, explain the procedure to her family.
- Obtain informed consent for starting an IV infusion.
- Start an IV infusion (two if the woman is in shock) using a large-bore (16-gauge or largest available) cannula or needle.
- Infuse IV fluids (normal saline or Ringer's lactate) at a rate appropriate for the woman's condition.
- **Note:** If the **woman is in shock**, avoid using plasma substitutes (e.g. dextran). There is no evidence that plasma substitutes are superior to normal saline in the resuscitation of a woman in shock, and dextran can be harmful in large doses.
- If a **peripheral vein cannot be cannulated**, perform a venous cutdown (**Fig. S-1, page S-4**).

BASIC PRINCIPLES FOR PROCEDURES

Before any simple (non-operative) procedure, the following steps are necessary:

- Where feasible, ensure that the woman has a companion of her choice during the procedure.
- Gather and prepare all supplies. A lack of needed supplies can disrupt a procedure.
- Explain the procedure and the need for it to the woman and any accompanying family members the woman would like to have involved in decision-making. If the woman is unconscious, explain the procedure to her family.
- Obtain informed consent for the procedure.
- Estimate the length of time for the procedure and provide pain medication accordingly (**page C-55**).

- Place the woman in a position appropriate for the procedure being performed. For example, when performing manual vacuum aspiration, use the lithotomy position; during cardiopulmonary resuscitation on a pregnant woman with gestational age of 20 weeks or more, perform left lateral tilt.
- Clean hands with soap and water (**page C-26**) or alcohol handrub (**page C-28**) and put on gloves appropriate for the procedure (**Table C-2, page C-31**).
- If the **vagina and cervix need to be prepared with an antiseptic** for the procedure (e.g. manual vacuum aspiration):
 - Wash the woman's lower abdomen and perineal area with soap and water, if necessary.
 - Gently insert a sterile speculum or retractor(s) into the vagina.
 - Apply antiseptic solution (e.g. iodophors, chlorhexidine) three times to the vagina and cervix using a sterile ring forceps and a cotton or gauze swab.
- If the **skin needs to be prepared with an antiseptic** for the procedure (e.g. for a caesarean birth):
 - Wash the area with soap and water, if necessary.
 - Apply antiseptic solution (e.g. iodophors, chlorhexidine) three times to the area using a sterile ring forceps and a cotton or gauze swab. If the **swab is held with a gloved hand**, do not contaminate the glove by touching unprepared skin.
 - Begin at the centre of the area and work outward in a circular motion away from the area.
 - At the edge of the sterile field, discard the swab.
 - Never go back to the middle of the prepared area with the same swab. Keep your arms and elbows high and surgical dress away from the surgical field.
- If the **procedure is conducted under local, paracervical or spinal anaesthesia**, keep the woman informed about progress during the procedure.
- After the procedure, inform the woman and accompanying family members of the woman's choice about:
 - how the procedure went;

- complications linked to the procedure (e.g. reaction to anaesthesia, too much bleeding, accidental injury);
 - any pertinent findings;
 - side effects to expect;
 - managing pain related to the procedure; and
 - estimated length of stay in the facility after the procedure.
- Provide instructions on post-procedure care and follow-up, including danger signs that indicate that the woman should return immediately to the facility for care.

CLINICAL USE OF BLOOD, BLOOD PRODUCTS, AND REPLACEMENT FLUIDS

C-37

Obstetric care sometimes requires blood transfusions. It is important to use blood, blood products and replacement fluids appropriately and to be aware of the principles designed to assist health workers in deciding when (and when not) to transfuse.

The appropriate use of blood products is defined as the transfusion of safe blood products to treat a condition leading to significant morbidity or mortality that cannot be prevented or managed effectively by other means.

Conditions that might require blood transfusion include:

- postpartum haemorrhage leading to shock;
- loss of a large volume of blood at operative birth; and
- severe anaemia, especially in later pregnancy or if accompanied by cardiac failure.

Note: For anaemia in early pregnancy, treat the cause of anaemia and provide haematinics.

District hospitals should be prepared for the urgent need for blood transfusion. It is mandatory for obstetric units to keep stored blood available, especially type O negative blood and fresh frozen plasma, as these can be life-saving.

UNNECESSARY USE OF BLOOD PRODUCTS

Used correctly, blood transfusions can save lives and improve health. Like any therapeutic intervention, however, transfusion can result in acute or delayed complications, and it carries the risk of transmission of infectious agents. It is also expensive and uses scarce resources.

- Transfusion is often unnecessary for the following reasons:
 - Conditions that might eventually require transfusion can often be prevented by early treatment or prevention programmes.
 - Transfusions of whole blood, red cells or plasma are often given to prepare a woman quickly for planned surgery or to allow earlier discharge from the hospital. Other treatments, such as the infusion of IV fluids, are often cheaper, safer and equally effective (**page C-34**).
- Unnecessary transfusion can:
 - expose the woman to unnecessary risks; and

- cause a shortage of blood products for women in real need. Blood is an expensive, scarce resource.

RISKS OF TRANSFUSION

Before prescribing blood or blood products for a woman, it is essential to consider the risks of transfusing against the risks of not transfusing.

WHOLE BLOOD OR RED CELL TRANSFUSION

- The transfusion of red blood cell products carries a risk of incompatible transfusion and serious haemolytic transfusion reactions.
- Blood products can transmit infectious agents—including HIV, hepatitis B, hepatitis C, syphilis, malaria and Chaga's disease—to the recipient.
- Any blood product can become bacterially contaminated and very dangerous if it is manufactured or stored incorrectly.

PLASMA TRANSFUSION

- Plasma can transmit most of the infections present in whole blood.
- Plasma can also cause transfusion reactions.
- There are very few clear indications for plasma transfusion (e.g. coagulopathy) and the risks often outweigh any possible benefit.

BLOOD SAFETY

- The risks associated with transfusion can be reduced by:
 - effective blood donor selection, deferral and exclusion;
 - screening for transfusion-transmissible infections in the blood donor population (e.g. HIV/AIDS and hepatitis);
 - quality-assurance programmes;
 - high-quality blood grouping, compatibility testing, component separation, and storage and transportation of blood products; and
 - appropriate clinical use of blood and blood products.

SCREENING FOR INFECTIOUS AGENTS

- Every unit of donated blood should be screened for transfusion-transmissible infections using the most appropriate and effective tests, in accordance with both national policies and the prevalence of infectious agents in the potential blood donor population.
- All donated blood should be screened for the following:
 - HIV-1 and HIV-2
 - hepatitis B surface antigen
 - *Treponema pallidum* antibody (syphilis).
- Where possible, all donated blood should also be screened for:
 - hepatitis C;
 - Chagas disease, in countries where the seroprevalence is significant;
 - malaria, in areas with high prevalence of malaria and in low-prevalence countries when donors have travelled to malarial areas.
- No blood or blood product should be released for transfusion until all nationally required tests are shown to be negative.
- Perform compatibility tests on all blood components transfused, even if, in life-threatening emergencies, the tests are performed after the blood products have been issued.

Blood that has not been obtained from appropriately selected donors and that has not been screened for transfusion-transmissible infectious agents (e.g. HIV, hepatitis) in accordance with national requirements should not be issued for transfusion, other than in the most exceptional life-threatening situations.

PRINCIPLES OF CLINICAL TRANSFUSION

The fundamental principle of appropriate use of blood or blood product is that transfusion is only one element of managing urgent care for women. When there is sudden rapid loss of blood due to haemorrhage, surgery or complications of childbirth, the most urgent need is usually the rapid replacement of the fluid lost from circulation.

Transfusion of red cells might also be vital to restoring the oxygen-carrying capacity of the blood.

Minimize “wastage” of a woman’s blood (to reduce the need for transfusion) by:

- using replacement fluids for resuscitation;
- minimizing the blood taken for laboratory use;
- using the best anaesthetic and surgical techniques to minimize blood loss during surgery;
- salvaging and reinfusing surgical blood lost during procedures (autotransfusion), where appropriate (**page S-17**).

Principles to remember:

- Transfusion is only one element of managing a woman’s care.
- Decisions about prescribing a transfusion should be based on national guidelines on the clinical use of blood, taking the woman’s needs into account.
- Blood loss should be minimized to reduce a woman’s need for transfusion.
- A woman with acute blood loss should receive effective resuscitation (IV replacement fluids, oxygen, etc.) while the need for transfusion is being assessed.
- The woman’s haemoglobin value, although important, should not be the sole deciding factor in starting the transfusion. The decision to transfuse should be supported by the need to relieve clinical signs and symptoms and prevent significant morbidity and mortality.
- The clinician should be aware of the risks of transfusion-transmissible infection in blood products that are available.
- Transfusion should be prescribed only when the benefits to the woman are likely to outweigh the risks.
- A trained person should monitor the transfused woman and respond immediately if any adverse effects occur (**page C-41**).
- The clinician should record the reason for transfusion and investigate any adverse effects (**page C-43**).

PRESCRIBING BLOOD

Prescribing decisions should be based on national guidelines on the clinical use of blood, taking the woman's needs into account.

- Before prescribing blood or blood products for a woman, keep in mind the following:
 - expected improvement in the woman's clinical condition;
 - methods to minimize blood loss to reduce the woman's need for transfusion;
 - alternative treatments that could be given, including IV replacement fluids or oxygen, before making the decision to transfuse;
 - specific clinical or laboratory indications for transfusion;
 - risks of transmitting HIV, hepatitis, syphilis or other infectious agents through the blood products that are available;
 - benefits of transfusion versus risk for the particular woman;
 - other treatment options if blood is not available in time;
 - the need for a trained person to monitor the woman and immediately respond if a transfusion reaction occurs.

MONITORING THE TRANSFUSED WOMAN

For each unit of blood transfused, monitor the woman at the following stages:

- before starting the transfusion;
- at the onset of the transfusion;
- 15 minutes after starting the transfusion;
- at least every hour during the transfusion;
- at four-hour intervals after completing the transfusion.

Closely monitor the woman during the first 15 minutes of the transfusion and regularly thereafter to detect early symptoms and signs of adverse effects.

At each of these stages, record the following information on the woman's chart:

- general appearance
- temperature
- pulse
- blood pressure
- respiration
- fluid balance (oral and IV fluid intake, urinary output).

In addition, record:

- the time the transfusion is started;
- the time the transfusion is completed;
- the volume and type of all products transfused;
- the unique donation numbers of all products transfused; and
- any adverse effects.

RESPONDING TO A TRANSFUSION REACTION

Transfusion reactions may range from a minor skin rash to anaphylactic shock. Stop the transfusion and keep the IV line open with IV fluids (normal saline or Ringer's lactate) while making an initial assessment of the acute transfusion reaction and seeking advice. If the **reaction is minor**, give promethazine 10 mg by mouth and observe.

MANAGING ANAPHYLACTIC SHOCK FROM MISMATCHED BLOOD TRANSFUSION

- Manage as for shock (**page S-1**) and give:
 - adrenaline 1:1000 solution (0.1 mL in 10 mL normal saline or Ringer's lactate) IV slowly;
 - promethazine 10 mg IV; and
 - hydrocortisone 1 g IV every two hours as needed.
- If **bronchospasm occurs**, give aminophylline 250 mg in 10 mL normal saline or Ringer's lactate IV slowly.
- Combine resuscitation measures above until stabilized.

- Monitor renal, pulmonary and cardiovascular functions.
- Transfer to referral centre when stable.

DOCUMENTING A TRANSFUSION REACTION

- Immediately after the reaction occurs, take the following samples and send with a request form to the blood bank for laboratory investigations:
 - immediate post-transfusion blood samples:
 - one clotted;
 - one anticoagulated (EDTA/Sequestrene) taken from the vein opposite the infusion site;
 - the blood unit and giving set containing red cell and plasma residues from the transfused donor blood;
 - the first specimen of the woman's urine following the reaction.
- If **septic shock is suspected due to a contaminated blood unit**, take a blood culture in a special blood culture bottle.
- Complete a transfusion reaction report form.
- After the initial investigation of the transfusion reaction, send the following to the blood bank for laboratory investigations:
 - blood samples at 12 hours and 24 hours after the start of the reaction:
 - one clotted;
 - one anticoagulated (EDTA/Sequestrene) taken from the vein opposite the infusion site;
 - all urine for at least 24 hours after the start of the reaction.
- Immediately report all acute transfusion reactions, with the exception of mild skin rashes, to a medical officer and to the blood bank that supplied the blood.
- Record the following information on the woman's chart:
 - type of transfusion reaction;
 - length of time after the start of transfusion that the reaction occurred;

- volume and type of blood products transfused; and
- unique donation numbers of all products transfused.

REPLACEMENT FLUIDS: SIMPLE SUBSTITUTES FOR TRANSFUSION

Only normal saline (sodium chloride 0.9%) or balanced salt solutions that have a similar concentration of sodium to plasma are effective replacement fluids. These should be available in all hospitals where IV replacement fluids are used.

Replacement fluids are used to replace abnormal losses of blood, plasma or other extracellular fluids by increasing the volume of the vascular compartment. They are used principally in:

- management of women with established hypovolaemia (e.g. haemorrhagic shock);
- maintenance of normovolaemia in women with on-going fluid losses (e.g. surgical blood loss).

INTRAVENOUS REPLACEMENT THERAPY

Intravenous replacement fluids are the first-line treatment for hypovolaemia. Initial treatment with these fluids may be life-saving and can provide some time to control bleeding and obtain blood for transfusion if it becomes necessary.

CRYSTALLOID FLUIDS

- Crystalloid replacement fluids:
 - contain a similar concentration of sodium to plasma;
 - cannot enter cells because the cell membrane is impermeable to sodium; and
 - pass from the vascular compartment to the extracellular space compartment (normally only a quarter of the volume of crystalloid infused remains in the vascular compartment).
- To restore circulating blood volume (intravascular volume), infuse crystalloids in a volume at least three times the volume lost.

Dextrose (glucose) solutions are poor replacement fluids. Do not use them to treat hypovolaemia unless there is no other alternative.

COLLOID FLUIDS

- Colloid solutions are composed of a suspension of particles that are larger than crystalloids. Colloids tend to remain in the blood where they mimic plasma proteins to maintain or raise the colloid osmotic pressure of blood.
- Colloids are usually given in a volume equal to the blood volume lost. In many conditions where the capillary permeability is increased (e.g. trauma, sepsis), leakage out of the circulation will occur and additional infusions will be necessary to maintain blood volume.

Points to remember:

- There is no evidence that colloid solutions (albumin, dextrans, gelatins, hydroxyethyl starch solutions) have advantages over normal saline or balanced salt solutions for resuscitation.
- There is evidence that colloid solutions may have an adverse effect on survival.
- Colloid solutions are much more expensive than normal saline and balanced salt solutions.
- Human plasma should not be used as a replacement fluid. All forms of plasma carry a risk, similar to that of whole blood, of transmitting infection, such as HIV and hepatitis.
- Plain water should never be infused intravenously. It will cause haemolysis and will probably be fatal.

There is a very limited role for colloids in resuscitation.

SAFETY

Before giving any IV infusion:

- check that the seal of the infusion bottle or bag is not broken;
- check the expiry date;
- check that the solution is clear and free of visible particles.

MAINTENANCE FLUID THERAPY

Maintenance fluids are crystalloid solutions, such as dextrose or dextrose in normal saline, used to replace normal physiological losses through skin, lungs, faeces and urine. If it is anticipated that the woman will receive IV fluids for 48 hours or more, infuse a balanced electrolyte solution (e.g. potassium chloride 1.5 g in 1 L IV fluids) with dextrose. The volume of maintenance fluids required by a woman will vary, particularly if the woman has fever or if the ambient temperature or humidity is high, in which case losses will increase.

OTHER ROUTES OF FLUID ADMINISTRATION

There are other routes of fluid administration in addition to the IV route.

ORAL AND NASOGASTRIC ADMINISTRATION

- This route can often be used for women who are mildly hypovolaemic and for women who can receive oral fluids.
- Oral and nasogastric administration should not be used if:
 - the woman is severely hypovolaemic;
 - the woman is unconscious;
 - there are gastrointestinal lesions or reduced gut motility (e.g. obstruction);
 - surgery with general anaesthesia is imminent.

RECTAL ADMINISTRATION

- Rectal administration of fluids is not suitable for severely hypovolaemic women.
- Advantages of rectal administration include the following:
 - It allows the ready absorption of fluids.
 - Absorption ceases and fluids are ejected when hydration is complete.
 - It is administered through a plastic or rubber enema tube inserted into the rectum and connected to a bag or bottle of fluid.
 - The fluid rate can be controlled by using an IV set, if necessary.

- The fluids do not have to be sterile. A safe and effective solution for rectal rehydration is 1 L of clean drinking water to which a teaspoon of table salt is added.

SUBCUTANEOUS ADMINISTRATION

- Subcutaneous administration can occasionally be used when other routes of administration are unavailable, but this method is unsuitable for severely hypovolaemic women.
- Sterile fluids are administered through a cannula or needle inserted into the subcutaneous tissue (the abdominal wall is a preferred site).

Solutions containing dextrose can cause tissue to die and should not be given subcutaneously.

This chapter briefly discusses the use of prophylactic antibiotics before an obstetrical procedure, therapeutic use of antibiotics for suspected or established severe pelvic infection, and management of antibiotic allergies.

- Prophylactic antibiotics are given to help prevent infection.
- **If a woman is suspected to have or is diagnosed as having an infection**, therapeutic antibiotics are indicated.

Recommendations for antibiotic treatment and prevention of specific conditions are discussed in the chapters on shock (S-1), operative care principles (C-65), normal labour and childbirth (C-77), vaginal bleeding in early pregnancy (S-7), vaginal bleeding after childbirth (S-29), fever during pregnancy and labour (S-113) and after childbirth (S-127), unsatisfactory progress of labour (S-71), difficulty in breathing (S-149), abdominal pain in early pregnancy (S-137), abdominal pain in later pregnancy and after childbirth (S-141), loss of fetal movements (S-155), prelabour rupture of membranes (S-159), induction and augmentation of labour (P-17), breech birth (P-45), caesarean birth (P-53), episiotomy (P-85), manual removal of placenta (P-91), repair of vaginal and perineal tears (P-97), correcting uterine inversion (P-111), repair of ruptured uterus (P-115), uterine and utero-ovarian artery ligation (P-119), postpartum hysterectomy (P-123), and salpingectomy for ectopic pregnancy (P-131).

Infection during pregnancy and the postpartum period can be caused by a combination of organisms, including aerobic and anaerobic cocci and bacilli. Antibiotics should be started based on specific indications, including:

- prevention of infection in the setting of established risk factors (e.g. vaginal colonization with Group B streptococcus);
- prophylaxis for medical procedures; and
- treatment of confirmed or suspected infection based on the clinical presentation of the woman.

Whenever possible, cultures and antibiotic sensitivities should be obtained (e.g. urine, vaginal discharge, pus) before initiating antibiotic treatment for a suspected infection so that treatment can be adjusted based on culture results or if there is no clinical response with treatment. However, prompt treatment of severe infections based on clinical presentation should not be delayed if a facility does not have the capacity to process cultures or to collect cultures in a timely manner.

If bacteraemia (presence of bacteria in the blood) or septicaemia (presence and multiplication of bacteria in the blood) is suspected, a blood culture should be done whenever feasible. Uterine infection can follow an abortion or

childbirth and is a major cause of maternal death. Broad spectrum antibiotics often are required to treat these infections. In cases of unsafe abortion and non-institutional births, antitetanus prophylaxis should also be provided as part of comprehensive management.

PROVIDING PROPHYLACTIC ANTIBIOTICS

Performing certain obstetrical procedures (e.g. caesarean birth, manual removal of placenta) increases a woman's risk of infectious morbidity. This risk can be reduced by:

- following recommended infection prevention and control practices (**page C-25**); and
- providing prophylactic antibiotics at the time of the procedure.

Whenever possible, give prophylactic intravenous antibiotics 15–60 minutes before the start of a procedure to achieve adequate blood levels of the antibiotic at the time of the procedure. One dose of prophylactic antibiotics is sufficient and is no less effective than three doses or 24 hours of antibiotics for preventing infection after an obstetrical procedure. If the **procedure lasts longer than six hours or blood loss is 1500 mL or more**, give a second dose of prophylactic antibiotics to maintain adequate blood levels during the procedure.

Obstetrical procedures for which antibiotic prophylaxis is recommended for the woman include the following:

- elective and emergency caesarean (**note:** prophylaxis to be given before starting the skin incision whenever possible) (**P-53**);
- suturing of third and fourth degree genital tears (**P-104**);
- manual removal of the placenta (**P-91**); and
- placement of uterine balloon tamponade (**S-35**).

PROVIDING THERAPEUTIC ANTIBIOTICS

- For initial treatment of **serious infections of the pelvic organs (e.g. uterus, fallopian tubes, ovaries) or upper urinary tract**, give a combination of antibiotics:
 - ampicillin 2 g IV every six hours;
 - PLUS gentamicin 5 mg/kg body weight IV every 24 hours.

Pain relief may be required during labour and is required during and after operative procedures. Analgesic drugs and methods of support during labour, local anaesthesia, general principles for using anaesthesia and analgesia, and postoperative analgesia are discussed in this chapter.

ANALGESIC DRUGS DURING LABOUR

- The perception of pain during labour depends greatly on a woman's emotional state. Supportive care during labour provides reassurance and decreases the perception of pain (**page C-86**).
- If a **woman is distressed by pain**, encourage her to walk around or assume any comfortable position. Encourage her companion to massage her back or sponge her face between contractions. Encourage the use of breathing techniques and allow the woman to take a warm bath or shower if she chooses. For most women, this is enough to cope with the pain of labour. If necessary, offer the woman:
 - morphine 0.1 mg/kg body weight IM every four hours as needed, informing her of the advantages and disadvantages (see below) and obtaining consent;
 - promethazine 25 mg IM or IV if vomiting occurs.

Barbiturates and sedatives should not be used to relieve anxiety in labour.

DANGER

If **morphine is given to the woman within four hours before she gives birth**, the baby may suffer from respiratory depression. Naloxone is the antidote.

Note: Do not administer naloxone to newborns whose mothers are suspected of having recently abused narcotic drugs.

- If there are **signs of respiratory depression** in the newborn, begin resuscitation immediately:
 - After vital signs have been established, give naloxone 0.1 mg/kg body weight IV to the newborn.
 - If the **infant has adequate peripheral circulation after successful resuscitation**, naloxone can be given IM. Repeated doses may be required to prevent recurrent respiratory depression.

- If there are **no signs of respiratory depression** in the newborn, but **morphine was given within four hours before birth**, observe the baby expectantly for signs of respiratory depression and treat as above if they occur.

PREMEDICATION WITH PROMETHAZINE AND DIAZEPAM

Premedication is required for procedures that last longer than 30 minutes. The dose must be adjusted to the weight and condition of the woman and to the condition of the fetus (when present). Inform the woman of the advantages and disadvantages, and obtain consent.

- Offer morphine 0.1 mg/kg body weight IM, informing the woman of the advantages and disadvantages (**page C-55**) and obtaining consent.
- Give diazepam in increments of 1 mg IV and wait at least two minutes before giving another increment. A safe and sufficient level of sedation has been achieved when the woman's upper eyelid droops and just covers the edge of the pupil.
 - Monitor the respiratory rate every minute. If the **respiratory rate falls below 10 breaths per minute**, stop administration of all sedative or analgesic drugs.
 - Monitor the fetal heart rate at least every 15 minutes. If the **fetal heart rate falls below 100 beats per minute**, stop administration of all sedative or analgesic drugs.

LOCAL ANAESTHESIA

Local anaesthesia (lidocaine with or without adrenaline) is used to infiltrate tissue and block the sensory nerves.

- Because a woman under local anaesthesia remains awake and alert during the procedure, it is especially important to ensure:
 - counselling to increase cooperation and minimize her fears;
 - good communication throughout the procedure as well as physical reassurance from the provider, if necessary;
 - time and patience, as local anaesthetics do not take effect immediately.

- The following conditions are required for the safe use of local anaesthesia:
 - All members of the operating team must be knowledgeable and experienced in the use of local anaesthetics.
 - Emergency drugs and equipment (suction, oxygen, resuscitation equipment) should be readily available and in usable condition, and all members of the operating team trained in their use.

LIDOCAINE

Lidocaine preparations are usually 2% or 1% and require dilution before use (**Box C-1**). For most obstetric procedures, the preparation is diluted to 0.5%, which gives the maximum effect with the least toxicity.

BOX C-1. Preparation of lidocaine 0.5% solution

Combine:

- lidocaine 2%, one part; and
- normal saline or sterile distilled water, three parts (do not use glucose solution as it increases the risk of infection).

Or combine:

- lidocaine 1%, one part; and
- normal saline or sterile distilled water, one part.

ADRENALINE

Adrenaline causes local vasoconstriction. Its use with lidocaine has the following advantages:

- less blood loss;
- longer effect of anaesthetic (usually one to two hours); and
- less risk of toxicity because of slower absorption into the general circulation.

If the **procedure requires a small surface to be anaesthetized or requires less than 40 mL of lidocaine**, adrenaline is not necessary. For larger surfaces, however, especially when more than 40 mL is needed, adrenaline is required to reduce the absorption rate and thereby reduce toxicity.

The best concentration of adrenaline is 1:200 000 (5 mcg/mL). This gives a maximum local effect with the least risk of toxicity from the adrenaline itself (**Table C-3**).

Note: It is critical to measure adrenaline carefully and accurately using a syringe such as a bacillus Calmette-Guérin (BCG) or insulin syringe. Mixtures must be prepared observing strict infection prevention practices (**page C-25**).

TABLE C-3. Formulas for preparing 0.5% lidocaine solutions containing 1:200 000 adrenaline

Desired Amount of Local Anaesthetic Needed	Normal Saline/ Lidocaine 2%	Normal Saline/ Lidocaine 1%	Adrenaline 1:1000
20 mL	15 mL/5 mL	10 mL/10 mL	0.1 mL
40 mL	30 mL/10 mL	20 mL/20 mL	0.2 mL
100 mL	75 mL/25 mL	50 mL/50 mL	0.5 mL
200 mL	150 mL/50 mL	100 mL/100 mL	1.0 mL

COMPLICATIONS

PREVENTION OF COMPLICATIONS

All local anaesthetic drugs are potentially toxic. Major complications from local anaesthesia, however, are extremely rare (**Table C-5, page C-59**).

The best way to avoid complications is to prevent them:

- Avoid using concentrations of lidocaine stronger than 0.5%.
- If **more than 40 mL of the anaesthetic solution is to be used**, add adrenaline to delay dispersion. Procedures that might require more than 40 mL of 0.5% lidocaine are caesarean and repair of extensive perineal tears.
- Use the lowest effective dose.
- Observe the maximum safe dose (**Table C-4, page C-59**). For an adult, this is 4 mg/kg body weight of lidocaine without adrenaline and 7 mg/kg body weight of lidocaine with adrenaline. The anaesthetic effect should last for at least two hours. Doses can be repeated if needed after two hours.

TABLE C-4. Maximum safe doses of local anaesthetic drugs

Drug	Maximum Dose (mg/kg of body weight)	Maximum Dose for 60 kg Adult (mg)
Lidocaine	4	240
Lidocaine + adrenaline 1:200 000 (5 mcg/mL)	7	420

- Inject slowly.
- Avoid accidental injection into a vessel. There are three ways of doing this:
 - Moving needle technique (preferred for tissue infiltration): The needle is constantly in motion while injecting; this makes it impossible for a substantial amount of solution to enter a vessel.
 - Plunger withdrawal technique (preferred for nerve block when considerable amounts are injected into one site): The syringe plunger is withdrawn before injecting; if blood appears, the needle is repositioned and attempted again.
 - Syringe withdrawal technique: The needle is inserted and the anaesthetic is injected as the syringe is being withdrawn.

To avoid lidocaine toxicity:

- Use a dilute solution.
- Add adrenaline when more than 40 mL will be used.
- Use the lowest effective dose.
- Observe the maximum dose.
- Avoid IV injection.

DIAGNOSIS OF LIDOCAINE ALLERGY AND TOXICITY**TABLE C-5. Symptoms and signs of lidocaine allergy and toxicity**

Allergy	Mild Toxicity	Severe Toxicity	Life-Threatening Toxicity (very rare)
• Shock • Redness of skin • Skin rash/hives • Bronchospasm • Vomiting • Serum sickness	• Numbness of lips and tongue • Metallic taste in mouth • Dizziness/light-headedness • Ringing in ears • Blurred vision	• Sleepiness • Disorientation • Muscle twitching and shivering • Slurred speech	• Tonic-clonic convulsions • Respiratory depression or arrest • Cardiac depression or arrest

MANAGEMENT OF LIDOCAINE ALLERGY

- Give adrenaline 1:1000, 0.5 mL IM, and repeat every 10 minutes if necessary.
- In acute situations, give hydrocortisone 100 mg IV every hour.
- To prevent recurrence, give diphenhydramine 50 mg IM or IV slowly, then 50 mg by mouth every six hours.
- Treat bronchospasm with aminophylline 250 mg in normal saline 10 mL IV slowly.
- Laryngeal oedema may require immediate tracheostomy.
- For shock, begin standard shock management (**page S-1**).
- Severe or recurrent signs may require corticosteroids (e.g. hydrocortisone IV 2 mg/kg body weight every four hours until condition improves). In chronic situations give prednisone 5 mg or prednisolone 10 mg by mouth every six hours until the condition improves.

MANAGEMENT OF LIDOCAINE TOXICITY

Symptoms and signs of toxicity (**Table C-5, page C-59**) should alert the practitioner to immediately stop injecting and prepare to treat severe and life-threatening side effects. If **symptoms and signs of mild toxicity are observed**, wait a few minutes to see if the symptoms subside, check vital signs, talk to the woman and then continue the procedure, if possible.

CONVULSIONS

- Turn the woman to her left side, insert an airway and aspirate secretions.
- Give oxygen at 6–8 L per minute by mask or nasal cannulae.
- Give diazepam 1–5 mg IV in 1-mg increments. Repeat if convulsions recur.
- **Note:** The use of diazepam to treat convulsions may cause respiratory depression.

RESPIRATORY ARREST

- If the **woman is not breathing**, assist ventilation using an Ambu bag and mask or via endotracheal tube; give oxygen at 4–6 L per minute.

CARDIAC ARREST

- Hyperventilate with oxygen.
- Perform cardiac massage.
- If the **woman has not yet given birth**, immediately perform a caesarean (**page P-53**) using general anaesthesia.
- Give adrenaline 1:10 000, 0.5 mL IV.

ADRENALINE TOXICITY

- Systemic adrenaline toxicity results from inadvertent or excessive amounts of IV administration and results in:
 - restlessness
 - sweating
 - hypertension
 - cerebral haemorrhage
 - rapid heart rate
 - ventricular fibrillation.
- Local adrenaline toxicity occurs when the concentration is excessive, and results in ischaemia at the infiltration site with poor healing.

GENERAL PRINCIPLES FOR ANAESTHESIA AND ANALGESIA

- The keys to pain management and comfort are:
 - supportive attention from staff before, during and after a procedure (helps reduce anxiety and lessen pain);
 - a provider who is comfortable working with women who are awake and who is trained to use instruments gently; and
 - the selection of an appropriate type and level of pain medication.

- Tips for performing procedures on women who are awake include the following:
 - Explain each step of the procedure before performing it.
 - Use adequate premedication in cases expected to last longer than 30 minutes.
 - Give analgesics or sedatives at an appropriate time before the procedure (30 minutes before for IM and 60 minutes before for oral medication) so that maximum relief will be provided during the procedure.
 - Use dilute solutions in adequate amounts.
 - Check the level of anaesthesia by pinching the area with forceps. If **the woman feels the pinch**, wait two minutes and then retest.
 - Wait a few seconds after performing each step or task to allow the woman to prepare for the next one.
 - Move slowly, without jerky or quick motions.
 - Handle tissue gently and avoid undue retraction, pulling or pressure.
 - Use instruments with confidence.
 - Avoid saying things like “this won’t hurt” if, in fact, it will hurt; or “I’m almost finished” if you are not almost finished.
 - Talk with the woman throughout the procedure.
- The need for supplemental analgesic or sedative medications (by mouth, IM or IV) depends on:
 - the emotional state of the woman;
 - the procedure to be performed (**Table C-6, page C-63**);
 - the anticipated length of the procedure; and
 - the skill of the provider and the assistance of the staff.

TABLE C-6. Analgesia and anaesthesia options

Procedure	Analgesia/Anaesthesia Options^a
Breech birth	• General methods of labour support (page C-85) • Pudendal block (page P-3)
Caesarean	• Spinal anaesthesia (page P-11) • Local anaesthesia (page P-7) • Ketamine (page P-13) • General anaesthesia
Cervical tears (extensive)	• Morphine and diazepam (pages C-55, C-56) • Ketamine (page P-13)
Colpotomy/ Culdocentesis	• Local anaesthesia (page C-56)
Craniotomy/ Craniocentesis	• Emotional support and encouragement (page C-9) • Diazepam (page C-56) • Pudendal block (page P-3)
Dilatation and curettage	• Paracervical block (page P-1) • Morphine (page C-55)
Episiotomy	• Local anaesthesia (page C-56) • Pudendal block (page P-3)
Forceps birth	• Emotional support and encouragement (page C-9) • Pudendal block (page P-3)
Labour and childbirth	• General methods of labour support (page C-85) • Morphine and promethazine (pages C-55, C-56)
Laparotomy	• General anaesthesia • Spinal anaesthesia (page P-11)
Manual removal of placenta	• Morphine and diazepam (pages C-55, C-56) • Ketamine (page P-13)
Manual vacuum aspiration	• Paracervical block (page P-1) • Morphine (page C-55)
Perineal tears (first and second degree)	• Local anaesthesia (page C-56) • Pudendal block (page P-3)
Perineal tears (third and fourth degree)	• Pudendal block (page P-3) • Ketamine (page P-13) • Local anaesthesia, morphine and diazepam (pages C-55, C-56) • Spinal anaesthesia (page P-11)
Uterine inversion (correction of)	• Morphine and diazepam (pages C-55, C-56) • General anaesthesia
Vacuum-assisted birth	• Emotional support and encouragement (page C-9) • Pudendal block (page P-3)

^aThe preferred analgesia/anaesthesia option is listed in bold.

POSTOPERATIVE ANALGESIA

Adequate postoperative pain control is important. A woman who is in severe pain does not recover well.

Note: Avoid over-sedation as this will limit mobility, which is important during the postoperative period.

Good postoperative pain control regimens include:

- non-narcotic mild analgesics, such as paracetamol 500 mg by mouth as needed;
- narcotics such as morphine 0.1 mg/kg body weight IM every four hours as needed, informing the woman of the advantages and disadvantages (**page C-55**) and obtaining consent;
- combinations of lower doses of narcotics with paracetamol.

Note: If the **woman is vomiting**, narcotics may be combined with anti-emetics such as promethazine 25 mg IM or IV every four hours as needed.

The woman is the primary focus of the physician/midwife and nurse during any procedure. The surgical or scrub nurse has her attention focused on the procedure and the needs of the physician/midwife performing the procedure.

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PREOPERATIVE CARE PRINCIPLES

PREPARING THE OPERATING THEATRE

Ensure that:

- the operating theatre is clean (it should be cleaned after every procedure);
- necessary supplies and anaesthesia equipment are available, including medications and an oxygen cylinder;
- emergency equipment, medication and supplies for both adult and newborn resuscitation are available, in working order and not past the expiry date;
- there is an adequate supply of theatre dress for the anticipated members of the surgical team;
- clean/sterile linens are available; and
- sterile surgical instruments and supplies (e.g. gloves, gauze, instruments) are available and not beyond expiry date.

PREPARING A WOMAN FOR A SURGICAL PROCEDURE

- Explain the procedure to be performed and its purpose to the woman and any accompanying family members the woman would like to be involved in decision-making. If the woman is unconscious, explain the procedure to her family.
- Obtain informed consent for the procedure.
- Ensure that all members of the operating, anaesthesia and newborn (if indicated) teams have been mobilized and the level of urgency of the surgery has been communicated.
- Ensure that the indication for the procedure is documented.
- Assist the woman and her family in preparing emotionally and psychologically for the procedure (**page C-9**).

- Review the woman's medical history:
 - Check for any possible allergies.
 - Ensure that the woman has received the complete antitetanus regimen, and give one dose of tetanus vaccine, if necessary.
- Review any potential anaesthetic risks (difficult airway), anticipated difficulty with the procedure, anticipated duration of the procedure and blood loss, and any specific concerns.
- Send a blood sample for haemoglobin (or haematocrit), type and screen, and order blood for possible transfusion. Do not delay transfusion if needed.
- Administer premedication appropriate for the anaesthesia used (**page C-56**).
- Give an antacid (sodium citrate 0.3% 30 mL or magnesium trisilicate 300 mg) to reduce stomach acid in case there is aspiration.
- Ensure prophylactic intravenous antibiotics have been ordered and given at least 15–60 minutes before surgery.

For caesarean birth, prophylactic antibiotics should be given prior to skin incision, rather than intra-operatively after umbilical cord clamping.

- Wash the area around the proposed incision site with soap and water, if necessary.
- Do not shave the woman's pubic hair as this increases the risk of wound infection. The hair may be trimmed, if necessary.
- Monitor and record vital signs (blood pressure, pulse, respiratory rate, temperature, oxygen saturation, and fetal heart rate, if applicable).
- Confirm that the indication for the procedure is still valid just before the start of the procedure, as this might have changed, rendering the need for the procedure obsolete (e.g. an emergency caesarean birth for fetal distress may have to be abandoned if the baby is found to have died before the procedure is started).

- Catheterize the bladder if necessary and monitor urine output.
- Ensure that all relevant information is passed on to other members of the team (doctor/midwife, nurse, anaesthetist, assistant and others).

INTRA-OPERATIVE CARE PRINCIPLES

POSITION

Place the woman in a position appropriate for the procedure to allow:

- the optimum exposure of the operative site;
- access for the anaesthetist;
- access for the nurse to take vital signs and monitor IV drugs and infusions;
- the safety of the woman, preventing injuries and maintaining circulation; and
- maintenance of the woman's dignity and modesty.

Note: If the **woman has not given birth**, tilt the operating table to the left or place a pillow or folded linen under her right lower back to decrease supine hypotension syndrome.

SURGICAL HANDSCRUB

- Remove all jewellery.
- Hold hands above the level of the elbow, wet hands thoroughly and apply soap (preferably an iodophore such as betadine).
- Begin at the fingertips and lather and wash, using a circular motion:
 - Wash between all fingers.
 - Move from the fingertips to the elbows of one hand and then repeat with the second hand.
 - Wash for three to five minutes.
- Rinse each arm separately, fingertips first, holding hands above the level of the elbows.

- Dry hands with a sterile towel, wiping from the fingertips to the elbows, or allow hands to air dry.
- Ensure that scrubbed hands do not come into contact with objects (e.g. equipment, protective gown) that are not sterile. If the **hands touch a contaminated surface**, repeat surgical handscrub.

PREPARING THE INCISION SITE

- Prepare the skin with an antiseptic (e.g. iodophors):
 - Apply antiseptic solution three times to the incision site using a sterile ring forceps and a cotton or gauze swab. If the **swab is held with a gloved hand**, do not contaminate the glove by touching unprepared skin.
 - Begin at the proposed incision site and work outward in a circular motion away from the incision site.
 - At the edge of the sterile field, discard the swab.
- Never go back to the middle of the prepared area with the same swab. Keep your arms and elbows high, and keep surgical dress away from the surgical field.
- Drape the woman immediately after the area is prepared to avoid contamination:
 - If the drape has a window, place the window directly over the incision site first.
 - Unfold the drape away from the incision site to avoid contamination.

MONITORING

Monitor the woman's condition regularly throughout the procedure.

- Monitor vital signs (blood pressure, pulse, and respiratory rate), level of consciousness and blood loss.
- Record the findings on a monitoring sheet to allow quick recognition if the woman's condition deteriorates.
- Maintain adequate hydration throughout surgery.

MANAGING PAIN

Maintain adequate pain management throughout the procedure (**page C-55**). Women who are comfortable during a procedure are less likely to move and cause injury to themselves. Pain management can include:

- emotional support and encouragement
- local anaesthesia
- regional anaesthesia (e.g. spinal)
- general anaesthesia.

ANTIBIOTICS

- Give prophylactic antibiotics before starting the procedure (**page C-50**).

MAKING THE INCISION

- When choosing the incision type, carefully consider the woman's past surgical history, the indication for surgery, and potential technical difficulties and complications. For example, for an obstructed labour case with Bandl's ring formation, a sub-umbilical midline incision may be preferable to a Pfannenstiel's incision.
- Make the incision only as large as necessary for the procedure.
- Make the incision with great care and proceed one layer at a time.

HANDLING TISSUE

- Handle tissue gently.
- When using clamps, close the clamp only one ratchet (click), when possible. This will minimize discomfort and reduce the amount of dead tissue that remains behind at the end of the procedure, thus decreasing the risk of infection.

HAEMOSTASIS

Haemostasis is the process in which blood is maintained in a fluid state and confined to the vascular system.

- Ensure haemostasis throughout the procedure to help maintain adequate visualization and to optimize outcomes.
- Women with obstetrical complications often have anaemia. Therefore, keep blood loss to a minimum.

INSTRUMENTS AND SHARPS

- Start and finish the procedure with a count of instruments, sharps and sponges:
 - Perform the count every time a body cavity (e.g. uterus) is closed;
 - Document in the woman's record that the surgical counts were correct.
- Use instruments, especially sharps, carefully to reduce the risk of injury (**page C-32**). Use "safe zones" when handling and passing instruments and sharps:
 - Use a pan such as a kidney basin to carry and pass sharp items, and pass suture needles on a needle holder.
 - Alternatively, pass the instrument with the handle, rather than the sharp end, pointing toward the receiver.

DRAINAGE

- Always leave an abdominal drain in place if:
 - bleeding persists after hysterectomy;
 - a clotting disorder is suspected;
 - infection is present or suspected; or
 - a bladder or ureteral injury is suspected or has been identified and repaired.
- A closed drainage system can be used, or a corrugated rubber drain can be placed through the abdominal wall or pouch of Douglas.
- Remove the drain once the infection has cleared or when no pus or blood-stained fluid has drained for 48 hours.

SUTURE

- Select the appropriate type and size of suture for the tissue (**Table C-7**). Sizes are reported by the number of “0”s:
 - A smaller suture has a greater number of “0”s (e.g. a 000 [3-0] suture is smaller than a 00 [2-0] suture); a suture labelled as “1” is larger in diameter than a “0” suture.
 - A suture that is too small will be weak and may break easily; a suture that is too large in diameter will tear through tissue.
- Refer to the section describing specific procedures for the recommended size and type of suture.

TABLE C-7. Recommended suture types

Suture Type	Tissue	Recommended Number of Knots
Plain catgut	Fallopian tube	3 ^a
Chromic catgut	Muscle	3 ^a
Polyglycolic	Muscle, fascia, skin	4
Nylon	Skin	6
Silk	Skin, bowel	3 ^a

^a These are natural sutures. Do not use more than three knots because this will abrade the suture and weaken the knot.

DRESSING

At the conclusion of surgery, cover the surgical wound with a sterile dressing (**page C-73**).

POSTOPERATIVE CARE PRINCIPLES

INITIAL CARE

- Place the woman in the recovery position:
 - Position the woman on her side with her head slightly extended to ensure a clear airway.
 - Place her upper arm in front of her body for easy access to check blood pressure.

- Place legs so that they are flexed, with the upper leg slightly more flexed than the lower to maintain balance.
- Assess the woman's condition immediately after the procedure:
 - Check vital signs (blood pressure, pulse, respiratory rate, temperature) and oxygen saturation every 15 minutes during the first hour, then every 30 minutes for the next hour.
 - Evaluate for airway obstruction, hypoxia, haemorrhage (internal and external), hypertension and/or hypotension, postoperative pain, hypothermia/shivering, vomiting and aspiration, and residual narcosis.
 - Assess the woman's level of consciousness every 15 minutes until she is alert.

Note: Ensure that the woman has constant supervision until conscious.

- Ensure that there is a clear airway and adequate ventilation.
- Transfuse if necessary (**page C-37**).
- If **vital signs become unstable** or if the **haematocrit continues to fall despite transfusion**, quickly return to the operating theatre because bleeding might be the cause.

GASTROINTESTINAL FUNCTION

Gastrointestinal function typically returns rapidly for obstetrical patients. For most uncomplicated procedures, bowel function should be normal within 12 hours of surgery.

- If the **surgical procedure was uncomplicated**, give the woman a liquid diet and advance to a regular diet as tolerated.
- If there were **signs of infection**, or if the **caesarean was for obstructed labour or uterine rupture**, wait until bowel sounds are heard before giving liquids.
- If the **woman is receiving IV fluids**, continue the fluids until she is taking liquids well.
- If you **anticipate that the woman will receive IV fluids for 48 hours or more**, infuse a balanced electrolyte solution (e.g. potassium chloride 1.5 g in 1 L IV fluids).

- If the **woman receives IV fluids for more than 48 hours**, monitor electrolytes every 48 hours. Prolonged infusion of IV fluids can alter electrolyte balance.
 - Whenever a woman is receiving IV fluids, monitor her intake and output closely.
- Ensure that the woman is eating a regular diet before discharge from the hospital.

DRESSING AND WOUND CARE

The dressing is a protective barrier against infection while a healing process known as “re-epithelialization” occurs. Keep the dressing on the wound for the first day after surgery to protect against infection while re-epithelialization occurs. Thereafter, a dressing is not necessary.

- **If blood or fluid is leaking through the initial dressing**, do not change the dressing:
 - Reinforce the dressing.
 - Monitor the amount of blood/fluid lost by outlining the blood stain on the dressing with a pen.
 - **If bleeding increases or the blood stain covers half the dressing or more**, remove the dressing and inspect the wound. Replace with another sterile dressing.
- If the **dressing comes loose**, reinforce with more tape rather than removing the dressing. This will help maintain the sterility of the dressing and reduce the risk of wound infection.
- Change the dressing using sterile technique.
- The wound should be clean and dry, without evidence of infection or seroma, before the woman’s discharge from the hospital.

ANALGESIA

Adequate postoperative pain control is important (**page C-55**). A woman who is in severe pain does not recover well.

Note: Avoid over-sedation as this will limit mobility, which is important during the postoperative period.

BLADDER CARE

A urinary catheter may be required for some procedures. Early catheter removal reduces the risk of infection and encourages the woman to walk.

- If the **urine is clear**, remove the catheter eight hours after surgery or after the first postoperative night.
- If the **urine is not clear**, leave the catheter in place until the urine is clear.
- Wait 48 hours after surgery before removing the catheter if there was:
 - uterine rupture
 - prolonged or obstructed labour
 - massive perineal oedema
 - puerperal sepsis with pelvic peritonitis.

Note: Ensure that the urine is clear before removing the catheter.

- If the **bladder was injured** (from either uterine rupture or during caesarean or laparotomy):
 - Leave the catheter in place for a minimum of seven days and until the urine is clear.
 - If the **woman is not currently receiving antibiotics**, give nitrofurantoin 100 mg by mouth once daily until the catheter is removed, for prophylaxis against cystitis.
 - If **there is a suspicion of a bladder injury** (e.g. the woman has had prolonged obstructed labour), consider leaving the catheter in for seven days.

ANTIBIOTICS

- If there were **signs of infection or the woman currently has a fever**, continue antibiotics until the woman is fever-free for 48 hours (**page C-49**).

SUTURE REMOVAL

Major support for abdominal incisions comes from the closure of the fascial layer. Remove skin sutures five days after surgery.

FEVER

- A fever (temperature of 38°C or more) that occurs postoperatively should be evaluated (**page S-127**).
- Ensure that the woman is fever-free for a minimum of 24 hours before discharge from the hospital.

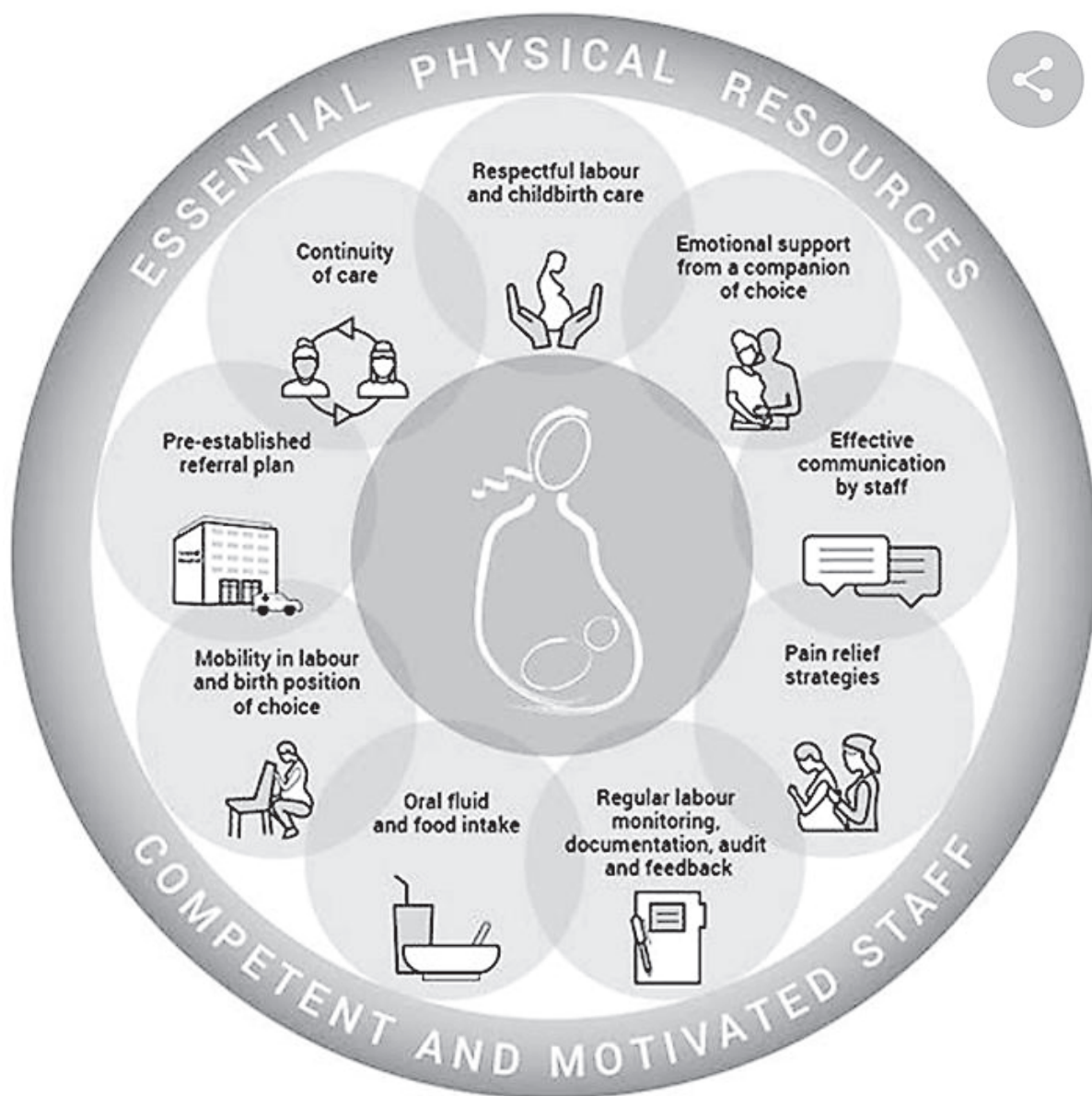
AMBULATION

Ambulation enhances circulation, encourages deep breathing and stimulates the return of normal gastrointestinal function. Encourage foot and leg exercises and mobilize as soon as possible, usually within 24 hours.

Encourage the woman to mobilize as soon as appropriate after giving birth, taking gentle exercise and making time to rest during the postnatal period.

SAFE SURGERY CHECKLIST

The World Health Organization (WHO) has undertaken a number of global and regional initiatives to address surgical safety, including the development of the *WHO Surgical Safety Checklist* (WHO, 2009). Countries are encouraged to adapt the checklist to their practice settings and circumstances using the checklist's accompanying implementation manual.



Every woman has a right to the highest attainable standard of health, the right to dignified, respectful health care throughout pregnancy and childbirth, and the right to be free from violence and discrimination. Abuse, neglect or disrespect during childbirth is a violation of a woman's fundamental human rights, as described in internationally adopted human rights standards and principles.

DIAGNOSIS AND CONFIRMATION OF LABOUR

- Suspect or anticipate labour if a woman has:
 - intermittent abdominal pain after 22 weeks of gestation;
 - abdominal pain/contractions often associated with blood-stained mucus discharge (show); and/or
 - watery vaginal discharge or a sudden gush of water.
- Confirm the onset of labour if there is:
 - cervical effacement—the progressive shortening and thinning of the cervix during labour; and
 - cervical dilatation—the increase in diameter of the cervical opening, measured in centimetres (**Fig. C-4, page C-81**).

DIAGNOSIS OF STAGE AND PHASE OF LABOUR

- Diagnose the stage of labour based on cervical dilatation (**Table C-8, page C-78**).
- Diagnose the phase of labour (**Table C-8, page C-78**) based on:
 - cervical dilatation during the first stage of labour; and
 - the woman's urge to push during the second stage of labour.

An incorrect diagnosis of labour can lead to unnecessary anxiety and interventions.

TABLE C-8. Diagnosis of stage and phase of labour based on cervical dilatation

Symptoms and Signs	Stage	Phase
• Cervix not dilated	False labour/ not in labour	
• Cervix dilated less than 5 cm	First	Latent
• Cervix dilated 5 cm up to 10 cm • Rate of dilatation typically 1 cm per hour or more • Fetal descent begins	First	Active
• Cervix fully dilated (10 cm)		Early
• Fetal descent continues • No urge to push	Second	(nonexpulsive)
• Cervix fully dilated (10 cm)		
• Presenting part of fetus reaches pelvic floor • Woman has the urge to push	Second	Late (expulsive)
The third stage of labour begins with the birth of the baby and ends with the expulsion of the placenta.		

INITIAL ASSESSMENT OF A WOMAN IN LABOUR

RAPID ASSESSMENT

- Ask the woman if she has had any danger signs (vaginal bleeding, gush of fluid or any problems such as fever, difficulty breathing, cough, leg pain, etc.) or has any concerns.
- Perform a rapid initial assessment (**page C-1**) of the woman (airway, breathing, circulation, vaginal bleeding, level of consciousness, convulsions, temperature higher than 38°C, abdominal pain).
- Assess fetal condition:
 - Listen to the fetal heart rate immediately after a contraction.
 - Count the fetal heart rate for a full minute.
 - If there are **fetal heart rate abnormalities** (less than 100 or more than 180 beats per minute), suspect fetal distress (**page S-109**).

- Respond immediately to any abnormalities in vital signs or other problems identified during the rapid initial assessment.

HISTORY

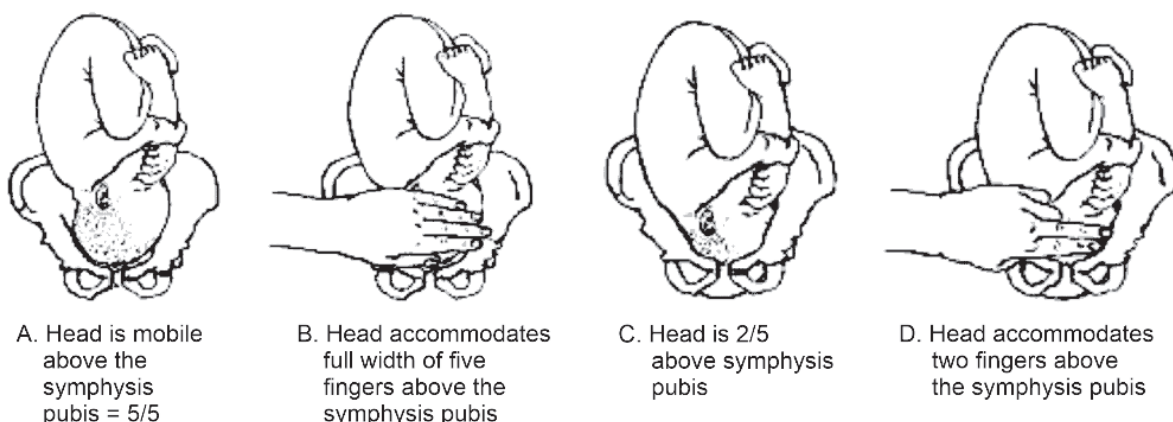
- Ask the woman about the history of this labour: approximate time contractions began and membranes ruptured.
- Ask the woman about or check her record for:
 - past obstetric, medical and surgical history, especially abdominal or pelvic surgeries or past caesarean(s);
 - problems during the current pregnancy (e.g. high blood pressure, pre-eclampsia, anaemia, malaria, sexually transmitted infections) and management;
 - haemoglobin;
 - tetanus vaccination status; and
 - syphilis, HIV and tuberculosis status.
- Check the woman's record. If she has no record:
 - Ask her when the birth is expected.
 - Determine the gestational age. If preterm (before 37 weeks of pregnancy are completed), manage as preterm labour (**page S-144**).

ABDOMINAL PALPATION

- Check the woman's abdomen for:
 - a caesarean scar;
 - a horizontal ridge across lower abdomen (if present, ask the woman to empty her bladder and observe her again).
- Assess uterine contractions:
 - Observe the woman's response to contractions:
 - Is she coping well or is she distressed?
 - Is she pushing or grunting? If so, manage as second stage of labour (**page C-97**).

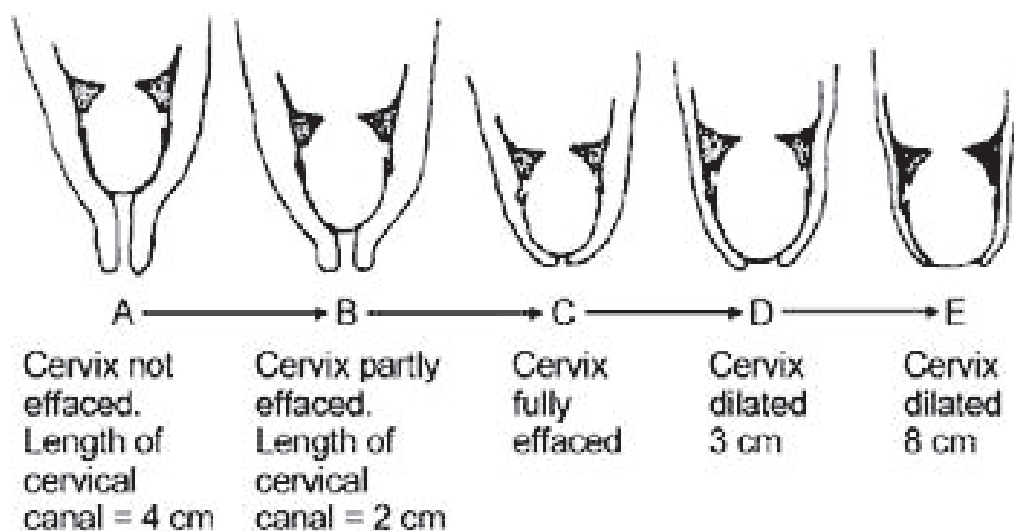
- Assess the number of uterine contractions in 10 minutes and their duration in seconds.
- Assess fetal lie—longitudinal or transverse—and fetal presentation.
 - Fetal lie other than longitudinal and presentations in labour other than cephalic are considered malpositions or malpresentations (page S-83).
- Assess **descent** in terms of fifths of fetal head palpable above the symphysis pubis (Fig. C-3 A–D):
 - A head that is entirely above the symphysis pubis is five fifths (5/5) palpable (Fig. C-3 A–B).
 - A head that is entirely below the symphysis pubis is zero fifths (0/5) palpable.

FIGURE C-3. Abdominal palpation for descent of the fetal head



VAGINAL EXAMINATION (page C-90)

- Assess cervical dilatation and effacement:

FIGURE C-4. Effacement and dilatation of the cervix

- Note the status of membranes. If the **membranes have ruptured**, note the colour of the draining amniotic fluid:
 - The presence of thick meconium indicates the need for close monitoring and possible intervention for management of fetal distress (**page S-109**).

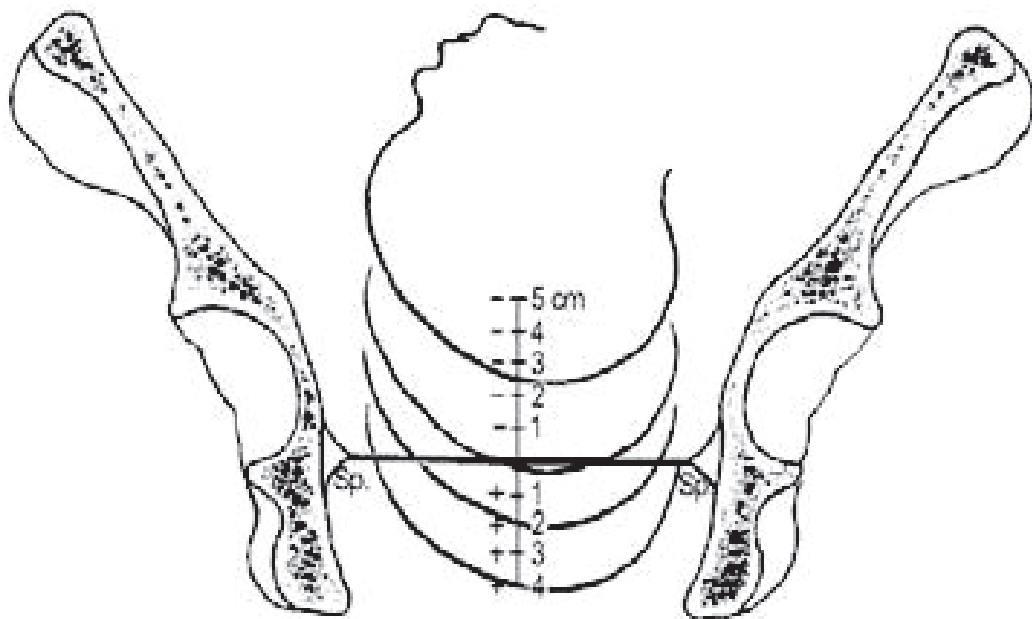
Routine antibiotics are **not recommended** in cases of meconium-stained amniotic fluid in the absence of other indications. Personnel experienced in neonatal resuscitation should attend the birth of infants if thick meconium liquor is noted, as meconium aspiration is clinically more significant in cases of thick meconium.

- The presence of foul-smelling watery discharge indicates the need to manage as amnionitis (**page S-163**).

Antibiotics should be administered when characteristics of the liquor suggest infection (**page S-182**).

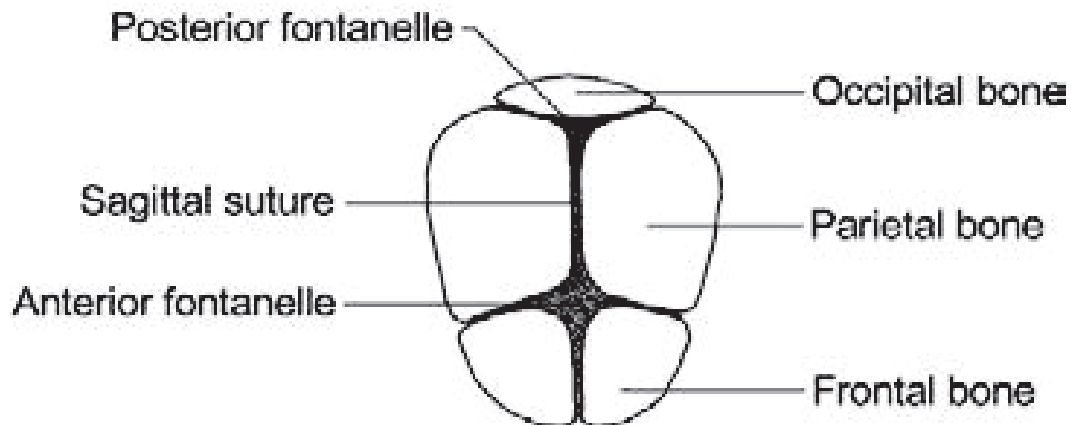
- An absence of fluid draining after rupture of the membranes is an indication of reduced volume of amniotic fluid, which may be associated with fetal distress.
- If necessary, a vaginal examination may be used to assess **descent** by relating the level of the fetal presenting part to the ischial spines of the maternal pelvis (**Fig. C-5, page C-82**).

FIGURE C-5. Assessing descent of the fetal head by vaginal examination; 0 station is at the level of the ischial spine (Sp)



Note: When there is a **significant degree of caput or moulding**, assessment by abdominal palpation using fifths of head palpable is more useful than assessment by vaginal exam.

- Determine **presenting part** during vaginal examination:
 - The most common presenting part is the vertex of the fetal head. If the **vertex is not the presenting part**, manage as a malpresentation (**Table S-16, page S-88**).
 - If the **vertex is the presenting part**, use landmarks on the fetal skull to determine the position of the fetal head in relation to the maternal pelvis (**Fig. C-6**).

FIGURE C-6. Landmarks of the fetal skull

- Determine the **position of the fetal head** during vaginal examination:
 - The fetal head normally engages in the maternal pelvis in an **occiput transverse position**, with the fetal occiput transverse in the maternal pelvis (**Fig. C-7**).

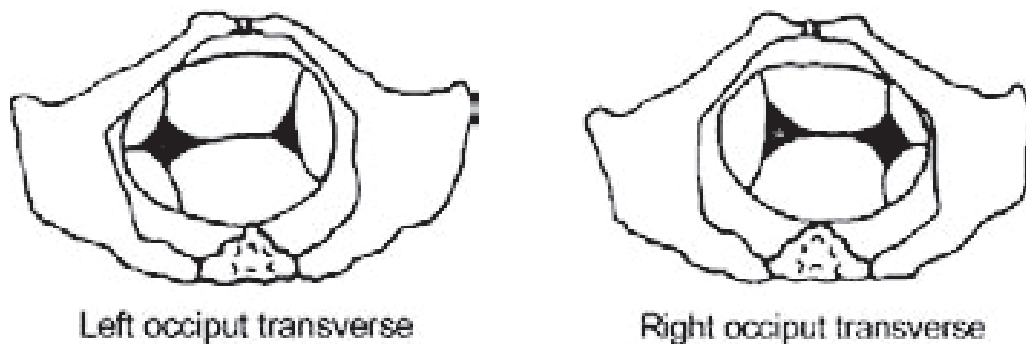
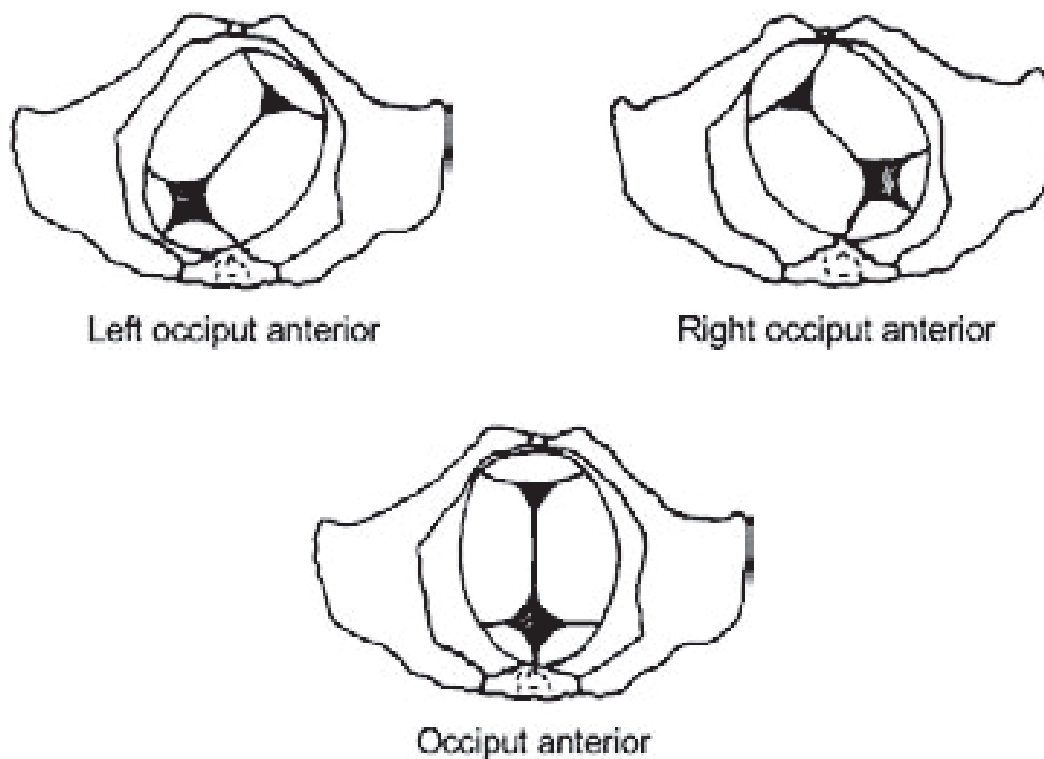
FIGURE C-7. Occiput transverse positions

FIGURE C-8. Occiput anterior positions

- With descent, the fetal head rotates so that the fetal occiput is anterior in the maternal pelvis (**occiput anterior position; Fig. C-8**). Failure of an occiput transverse position to rotate to an occiput anterior position should be managed as an occiput posterior position (**page S-91**).
- An additional feature of a normal presentation is a **well-flexed vertex (Fig. C-9, page C-85)**, with the occiput lower in the vagina than the sinciput.

- Assist the woman in assuming the position of her choice, and maintain eye contact during the examination to decrease any feelings of vulnerability.
- After the examination, provide information on your findings and how they affect the plan of care.
- At each vaginal examination, record the following:
 - status of membranes;
 - colour of amniotic fluid (liquor), if membranes are ruptured;
 - cervical dilatation and effacement; and
 - descent (can also be assessed abdominally).
- If the **cervix is not dilated on first examination**, it might not be possible to diagnose labour.
 - If **contractions persist**, re-examine the woman for cervical changes after four hours. At this stage, if there is **effacement and dilatation**, the woman is in labour; if there is **no change**, the diagnosis is false labour.
 - In the second stage of labour, perform vaginal examinations once every hour.

USING THE LABOUR CARE GUIDE (LCG)

The WHO Partograph has been replaced by the WHO Labour Care Guide, designed for the care of **All** women and their babies during labour and childbirth, at all levels of Healthcare. It includes assessments and observations that are essential for the care of all pregnant women, regardless of their risk status. Women at high risk of developing labour complications may require additional specialized monitoring and care. The principal aims of the LCG are to:

- Guide the monitoring and documentation of the well-being of women and babies and the progress of labour
- Guide skilled health personnel to offer supportive care throughout labour to ensure a positive childbirth experience for women
- Promptly identify & address potential complications, by providing **reference thresholds** for labour observations, to facilitate specific action(s) and timely referral if required
- Prevent unnecessary use of interventions in labour
- Support audit and quality improvement of labour management.

When to start using the LCG: plotting on the LCG begins in the active phase when the cervix is **5 cm** or more dilated. A sample LCG is included (**Fig. C-12, page C-95**). Note that the Labour Care Guide should be enlarged to full size before use. Record the following on it:

SECTION 1: Identifying information and labour characteristics at admission: Record the woman's name, gravida, para, hospital number, date and time of admission, and time of ruptured membranes or time elapsed since rupture of membranes (if rupture occurred before charting on the LCG began); time of active first stage diagnosis; risk factors

SECTION 2: Supportive Care

Respectful maternity care (RMC) is a fundamental human right of **all** pregnant women and is a core component of the WHO intrapartum care recommendations.

A companion of choice is recommended for all women throughout labour and childbirth. **Y**=Yes; **N**= No

Pain Relief: depends on the labouring woman's choice and includes Relaxation and manual techniques, parental opioids and epidural analgesia for healthy pregnant women. **Y**= Yes ; **N**=No : **D** = Declined

Oral Fluid : light diet & drink during labour allowed. **Y**= Yes ; **N**=No : **D** = Declined

Posture: mobility and a birth position of the individual woman's choice, including an **upright position**, is recommended **SP** + Supine ; **MO** =Mobile.

SECTION 3: Care of the Baby

Intermittent auscultation of the fetal heart rate with either a Doppler ultrasound or Pinard stethoscope is recommended for healthy pregnant women. Continuous cardiotocography is not recommended in healthy women undergoing spontaneous labour

Fetal Heart Rate: Auscultate every 15–30 minutes in the active first stage of labour, and every 5 minutes in the second stage of labour.

Baseline FHR (Alerts <110 , >160)

FHR Deceleration : **Y**=Yes; **N**= No, **E**= early, **L**=late, **V**= variable; (Alert Late deceleration)

Amniotic fluid: Record the colour of the amniotic fluid and the status of membranes at every vaginal examination: **I**: membranes intact, **C**: membranes ruptured, clear fluid; **M**: meconium-stained fluid + to +++; **B**: blood-stained fluid. (Alert **M+++** (thick meconium, **B**))

Fetal Position : **A**= occiput Anterior; **P**= occiput Posterior; **T**= occiput Transverse (Alert **P**, **T**)

Caput : grade from **0** to +++ (**Alert** +++)

Moulding: + = sutures apposed; ++ = sutures overlapped but reducible
+++ = sutures overlapped and not reducible. Note development of **caput succedaneum** (**Alert** +++)

Section 4: Care of the woman

Maternal Pulse Pulse: Record every 4 hours. (**Alert** < 60 , >120 bpm)

Blood pressure: Record every 4 hours. (**Alert** SBP <80, >140; DBP >90)

Temperature: Record every 4 hours. (**Alert:** <35.0, >37.5)

urinary output: Record when urine is passed Protein, Acetone and volume:
P= protein, A= acetone (**Alert** P++, A++)

Section 5: Labour Progress

Systematic intermittent monitoring of labour progression parameters is noted. For women with spontaneous labour onset, **the cervical dilatation rate threshold of 1 cm/hour during active first stage is inaccurate** to identify women at risk of adverse birth outcomes, and is not recommended

Labour may not naturally accelerate until a cervical dilatation threshold
The **duration of active first stage** (from 5 cm until full cervical dilatation) usually does not extend beyond **12 hours** in first labours, and usually does not extend beyond **10 hours** in subsequent labours

In first labours, birth is usually completed within **3 hours**, whereas in subsequent labours, birth is usually completed within **2 hours**.

Labour progress is recorded on the LCG once 5 cm cervical dilatation is reached by regular observation of:

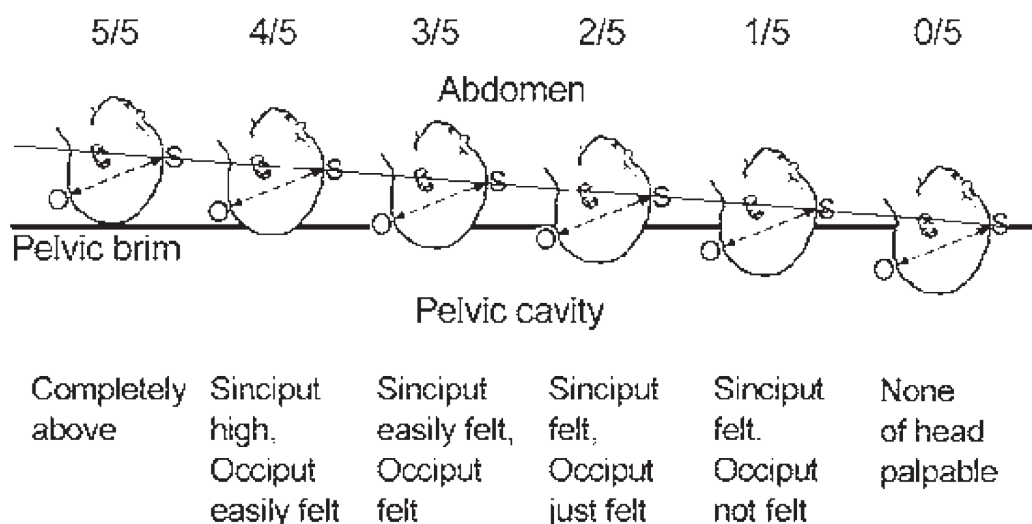
Contractions frequency per 10 min: record absolute number of contractions.
(**Alert** : <2 >5)

Duration of contractions in seconds (**Alert** <20, >60)

Cervical dilatation: Assess at every vaginal examination and mark a cross

(X) on the LCG. Begin plotting on the LCG at 5 cm.

Descent of the head: assessed by abdominal palpation: Refers to the part of the head (divided into five parts) palpable above the symphysis pubis; record as a circle (O) at every abdominal examination. At 0/5, the sinciput (S) is at the level of the symphysis pubis.

FIGURE C-11. Descent assessed by abdominal palpation**Section 6: Medication**

Oxytocin: Record the amount of oxytocin in units per litre (U/L) and in drops per minute every 30 minutes when used.

Other Medicine: Record any additional drugs given.

IV Fluids : record Y= yes, N=No

Section 7: Shared Decision Making

Aims to facilitate continuous communication with the woman and her companion, and consistent recording of all assessments and plans agreed. WHO recommends effective communication between maternity health providers and women in labour, including the use of simple and culturally appropriate language

Assessment: record overall assessment & any additional findings not previously documented

Plan: e.g continuation of routine monitoring, diagnostic tests, augmentation with Oxytocin, ARM, assisted birth with Vacuum/ forceps, Caesarean section. Women should be involved in discussions & decisions. Record the plan each time a clinical assessment is done based on a shared plan

FIGURE C-12. The Labour Care Guide (LCG)

WHO LABOUR CARE GUIDE

Name _____ Parity _____ Labour onset _____ Active labour diagnosis [Date _____]

Ruptured membranes [Date _____ Time _____] Risk factors _____

		Time	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
		Hours															
		ALERT	ACTIVE FIRST STAGE												SECOND STAGE		
SUPPORTIVE CARE	Companion	N															
	Pain relief	N															
	Oral fluid	N															
	Posture	SP															
BABY	Baseline FHR	< 110, ≥ 160															
	FHR deceleration	L															
	Amniotic fluid	M+++, B															
	Fetal position	P, T															
	Caput	+++															
	Moulding	+++															
WOMAN	Pulse	< 60, ≥ 120															
	Systolic BP	< 80, ≥ 140															
	Diastolic BP	≥ 90															
	Temperature °C	< 35.0, ≥ 37.5															
	Urine	P++, A++															
LABOUR PROGRESS	Contractions per 10 min	≤ 2, > 5															
	Duration of contractions	< 20, > 60															
	Cervix [Plot X]	10															
		9	≥ 2h														
		8	≥ 2.5h														
		7	≥ 3h														
		6	≥ 5h														
	Descent [Plot O]	5	≥ 6h														
		5															
		4															
3																	
2																	
MEDICATION	Oxytocin (U/L, drops/min)																
	Medicine																
	IV fluids																
	SHARED DECISION-MAKING	ASSESSMENT															
PLAN																	
INITIALS																	

INSTRUCTIONS: CIRCLE ANY OBSERVATION MEETING THE CRITERIA IN THE 'ALERT' COLUMN, ALERT THE SENIOR MIDWIFE OR DOCTOR AND RECORD THE ASSESSMENT AND ACTION TAKEN. IF, LABOUR EXTENDS BEYOND 12 h, PLEASE CONTINUE ON A NEW LABOUR CARE GUIDE.

Abbreviations: Y – Yes, N – No, D – Declined, U – Unknown, SP – Supine, MO – Mobile, E – Early, L – Late, V – Variable, I – Intact, C – Clear, M – Meconium, B – Blood, A – Anterior, P – Posterior, T – Transverse, P+ – Protein, A+ – Acetone

Fig. C-13, page C-96, is a sample Labour Care Guide for normal labour:

Case Study: Date: June 07, 2020 Time 06:00

Mary Jane (M J, 4th Gravida with 38 wks uncomplicated pregnancy, Para 2+1, presented with contractions and ruptured membranes for 1 hour. She previously had 2 normal deliveries, one of a live baby and one of a stillbirth at term, & a spontaneous miscarriage. She is taking prophylactic oral iron tablets. On examination she has regular contractions (3 every 10 minutes), cervix is 5 cm dilated and membranes already ruptured draining clear liquor. She is admitted to the labour ward & monitored by the midwife; She has no support person. She is having painful uterine contractions, 3 every 10 minutes, of moderate intensity, lasting 40 seconds, and requests pain relief. She drank some fruit juice & is walking.

Mary Jane chose her sister for support. The midwife instructed her about when and how to call for assistance. They also received instructions on relaxation techniques for pain relief. Another woman was in labour in the same room, the midwife placed a screen between the beds to provide privacy.

The baby moves, and FHR is 140 beats per minute (bpm), no decelerations

Her Pulse is 88 bpm, BP 120/80 mmHg, temperature 36.5°C.

She passed urine at admission, no protein or acetone detected.

Vaginal examination on admission shows Cervix is **5 cm** dilated, cephalic presentation. There is no caput or moulding and the fetal position is occiput posterior. Amniotic fluid is clear. Fetal descent: head is 4/5 palpable per abdomen.

All other clinical parameters are normal and M J is coping with labour. The midwife assesses the number and duration of uterine contractions half-hourly & plans to reassess cervical dilatation & clinical observations in 4 hours unless otherwise indicated.

Time 06:30: FHR 136 bpm without decelerations

Time 07:00: MJ is with her sister, using relaxation techniques for pain relief; drinking water when thirsty, and lying in bed in supine position. FHR 132 bpm with variable decelerations

Time 07:30: FHR 148 bpm no decelerations. M.J's pad is checked, the amniotic fluid is clear.

All other clinical parameters are normal & M J is coping well with labour, her midwife continues checking the FHR every 30 minutes and will check the amniotic fluid in the next vaginal examination

Time 10:00: Pulse 96 bpm, BP 128/84 mmHg, temp 36.9°C. She passed urine again, with no protein or acetone detected. M J complains of strong pains. Her sister left the labour ward & M J is alone, lying in bed in a supine position. Her pulse is 96 bpm, BP 128/84 mmHg, FHR is 151 bpm with variable decelerations. M J has 3 strong uterine contractions in 10 minutes, lasting 50 seconds each. Fetal head is 3/5 palpable per abdomen. Cervical dilatation is 8 cm and the fetal position is occiput transverse. Amniotic fluid shows meconium 1+. The midwife offers her a companion of her choice. M J wants to be accompanied by her sister who had left to speak with the family in the waiting room. The midwife gives directions to M J's sister on how to support M J & comfort her by using a cool, damp cloth on her face and body, & by massaging her back.

Time 13:00: M J has 3 uterine contractions in 10 minutes, lasting 50 seconds each. Fetal Head (descent) is 2/5 palpable above symphysis pubis. Cervical dilatation is 10 cm and the fetal position is occiput anterior. Amniotic fluid shows meconium 1+. FHR 132 bpm, without decelerations.

Time 13:30: Mary Jane has 4 uterine contractions in 10 minutes, lasting 50 seconds each. Fetal Head (Descent) is 0/5 palpable. FHR 118 bpm, with early decelerations.

Time 13:45: Childbirth takes place vaginally

Mary Jane had normal progress of labour and childbirth. During labour, she was encouraged to walk and to have a companion of her choice present. Clinical parameters remained within normal limits. Consequently, additional interventions were not required.

FIGURE C-13. Sample Labour Care Guide for normal labour

WHO LABOUR CARE GUIDE

Name: Mary Jane Williams Parity: 2 Labour onset: spontaneous Active labour diagnosis (Date: 06/07/20 |

Ruptured membranes (Date: 06/07/20 Time: 5:00 | Risk factors: History of stillbirth, anaemia

		Time	6:00	7:00	8:00	9:00	10:00	11:00	12:00	13:00	:	:	:	:	13:05	13:45	:	
		Hours	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
		ALERT	ACTIVE FIRST STAGE												SECOND STAGE			
SUPPORTIVE CARE	Companion	N	(N)	Y	Y	Y	(N)	Y	Y	Y					Y			
	Pain relief	N	(N)	Y	Y	Y	(N)	Y	Y	Y					Y			
	Oral fluid	N	Y	Y	Y	D	Y	Y	D	Y					Y			
	Posture	SP	MO	(SP)	MO	MO	(SP)	MO	MO	(SP)					(SP)			
BABY	Baseline FHR	<110, ≥160	110	136	132	148	133	145	138	128	151	133	144	125	153	120	132	
	FHR deceleration	L	N	N	V	N	N	N	N	N	V	N	N	N	N	N	N	
	Amniotic fluid	M+++; B	C								+					+		
	Fetal position	P, T	(P)								(T)					A		
	Caput	+++	0								+					+		
	Moulding	+++	0								+					++		
WOMAN	Pulse	<60, ≥120	88							96								
	Systolic BP	<80, ≥140	120							128								
	Diastolic BP	≥90	80							84								
	Temperature °C	<35.0, ≥37.5	36.5							36.9								
	Urine	P++, A++	-/-							-/-								
LABOUR PROGRESS	Contractions per 10 min	≥2, >5	3	3	3	3	3	3	3	3	3	3	4	3	3	3		
	Duration of contractions	<20, >60	40	40	40	40	40	45	40	45	50	50	50	40	50	50	50	
	Cervix (Plot X)	10																
		9	≈ 2h															
		8	≈ 2.5h															
		7	≈ 3h															
		6	≈ 5h															
	5	≈ 6h	x															
	Descent (Plot O)	5																
		4		0														
3																		
2															0			
1																		
0																		
MEDICATION	Oxytocin (μL, drops/min)	N	N	N	N	N	N	N	N	N								
	Medicine	N	N	N	N	N	N	N	N	N								
	IV fluids	N	N	N	N	N	N	N	N	N								
SHARED DECISION-MAKING	ASSESSMENT		PAIN RELIEF REQUIRED	NORMAL PROGRESS	NORMAL PROGRESS	NORMAL PROGRESS	PAIN RELIEF REQUIRED	NORMAL PROGRESS	NORMAL PROGRESS	NORMAL PROGRESS					NORMAL PROGRESS			
	PLAN		Offer companionship and relaxation techniques; continuous of routine monitoring	Continuation of routine monitoring	Continuation of routine monitoring	Continuation of routine monitoring	Offer companionship and maternal pain relief; encourage mobilisation; continuous monitoring	Continuation of routine monitoring	Continuation of routine monitoring	Continuation of routine monitoring					Continuation of routine monitoring			
INITIALS		LA	LA	LA	GP	GP	GP	GP	GP	GP					GP			

INSTRUCTIONS: CIRCLE ANY OBSERVATION MEETING THE CRITERIA IN THE 'ALERT' COLUMN, ALERT THE SENIOR MIDWIFE OR DOCTOR AND RECORD THE ASSESSMENT AND ACTION TAKEN IF LABOUR EXTENDS BEYOND 12H. PLEASE CONTINUE ON A NEW LABOUR CARE GUIDE.

Abbreviations: Y – Yes, N – No, D – Declined, U – Unknown, SP – Supine, MO – Mobile, E – Early L – Late, V – Variable, I – Intact, C – Clear, M – Meconium, B – Blood, A – Anterior, P – Posterior, T – Transverse, P+ – Protein, A+ – Acetone

PROGRESS OF FIRST STAGE OF LABOUR

- The following findings are suggestive of **satisfactory progress** in the first stage of labour:
 - regular contractions of progressively increasing frequency and duration;
 - rate of cervical dilatation within the Alert range at each vaginal examination during the active phase of labour
 - cervix well applied to the presenting part.
- The following findings are suggestive of **unsatisfactory progress** in the first stage of labour:
 - woman still in latent phase after eight hours of regular contractions;
 - OR irregular and infrequent contractions (two contractions or fewer in 10 minutes, each lasting less than 40 seconds) once active phase begins;
 - OR rate of cervical dilatation slower than the range give in the Alert column
 - OR cervix poorly applied to the presenting part;
 - OR secondary arrest of cervical dilatation and descent of presenting part in the presence of adequate frequency and strength of contractions.

Unsatisfactory progress in labour can lead to prolonged labour and must be recognized and addressed promptly to reduce the risk of ruptured uterus and other complications (**Table S-15, page S-74**).

PROGRESS OF SECOND STAGE OF LABOUR

- Findings suggestive of **satisfactory progress** in the second stage of labour are:
 - steady descent of fetus through birth canal; and
 - onset of expulsive (pushing) phase.
- Findings suggestive of **unsatisfactory progress** in the second stage of labour are:

- lack of descent of fetus through the birth canal;
- failure of expulsive effort to develop during the late (expulsive) phase.

NORMAL CHILDBIRTH

Prepare for birth by preparing all equipment and supplies and, if feasible, calling for an assistant.

- Ensure that the birthing room is warm (25°C), with no draughts.
- Never leave the woman alone.
- Ensure that the woman's bladder is empty (only catheterize the woman if the bladder is full and she is unable to void).
- Assist the woman into a position of her choice, as upright as possible to facilitate physiologic birth, and ensure her comfort in this position.
- Offer her emotional and physical support.
- Once the cervix is fully dilated and the woman is in the expulsive phase of the second stage (woman has the urge to push), encourage her to assume the position she prefers (**Fig. C-14, page C-99**).

FIGURE C-14. Some positions that a woman may adopt during childbirth



General methods of supportive care during labour are most useful in helping the woman tolerate labour pains.

- Support the woman to push as she wishes with contractions. **Do not** urge her to push, particularly if the fetus remains at the pelvic midpoint.
- Continue to monitor the woman's condition (mood and behaviour), the condition of the fetus (fetal heart rate) and progress of labour (frequency, intensity and duration of contractions; perineum thinning and bulging; visible descent of the fetal head during contractions).
- If, after 30 minutes of spontaneous expulsive efforts, the perineum does not begin to thin and stretch with contractions, do a vaginal examination to confirm full dilatation of the cervix.

Note: Episiotomy is **not** recommended as a routine procedure. There is no evidence that routine episiotomy decreases perineal damage, future vaginal prolapse or urinary incontinence. In fact, routine episiotomy is associated with an increase of third and fourth degree tears and subsequent anal sphincter muscle dysfunction.

BIRTH OF THE HEAD

- Ask the woman to pant or give only small pushes with contractions as the baby's head is born.
- To control the birth of the head, place the fingers of one hand against the baby's head to keep it flexed (bent).
- Continue to gently support the perineum as the baby's head is born.
- Once the baby's head is born, ask the woman not to push.

Note: Routine intrapartum nasal or oral suction should **not** be done, even in babies born through liquor with meconium.

- Feel around the baby's neck for the umbilical cord:
 - If the **cord is around the neck but it is loose**, slip it over the baby's head.
 - If the **cord is tight around the neck**, doubly clamp and cut it before unwinding it from around the neck.

COMPLETION OF THE BIRTH

- Allow the baby's head to turn spontaneously.
- After the head turns, place a hand on each side of the baby's head. Tell the woman to push gently with the next contraction.
- Reduce tears by supporting the birth of one shoulder at a time. Move the baby's head posteriorly to facilitate birth of the anterior shoulder.

Note: If there is **difficulty delivering the shoulders**, or if the infant's head retracts against the perineum as it is born, suspect shoulder dystocia (**page S-99**).

- Lift the baby's head anteriorly to deliver the shoulder that is posterior.
- Support the rest of the baby's body with one hand as it slides out.
- Place the baby on the mother's abdomen.
- Thoroughly dry the baby and wipe the eyes. Remove the wet cloth.

Note: For neonates born through meconium-stained amniotic fluid who start breathing on their own, suctioning of the mouth or nose or tracheal suctioning should not be performed.

- Note the time of birth.
- Keep the baby warm; position skin-to-skin with the mother and cover the baby's head and body.

- Assess the baby's breathing:

Note: Most babies begin crying or breathing spontaneously within 30 seconds of birth:

- If the **baby is crying or breathing** (chest rising at least 30 times per minute), leave the baby with the mother.
- If the **baby does not start breathing**, stimulate the baby by rubbing the back two or three times.
- If, after drying and brief stimulation, the baby is still not crying or breathing, **call for help**, clamp and cut the umbilical cord and move the newborn to a flat surface to initiate resuscitation, including initiating positive-pressure ventilation within one minute after birth (**page S-167**).

Anticipate the need for resuscitation and have a plan to get assistance for every baby, but especially if the mother has a history of eclampsia, bleeding, prolonged or obstructed labour, preterm birth, or infection.

- If the **baby is breathing normally**, clamp and cut the umbilical cord one to three minutes after the birth of the baby, while initiating simultaneous essential newborn care.

Note: Only clamp the cord early (within one minute) if the newborn needs to be moved immediately for resuscitation.

- Ensure that the baby is kept warm and in skin-to-skin contact on the mother's chest. Keep the baby covered with a soft, dry cloth or blanket, and ensure that the baby's head is covered to prevent heat loss.
- If the **mother is not well**, request the support of an assistant so that both mother and baby can be appropriately monitored and cared for.
- Palpate the woman's abdomen to rule out the presence of an additional baby(s), and proceed with active management of the third stage of labour.

ACTIVE MANAGEMENT OF THE THIRD STAGE OF LABOUR

Active management of the third stage (active delivery of the placenta) helps prevent postpartum haemorrhage. It includes administration of a uterotonic medication within one minute after the birth of the baby. It may also include:

- controlled cord traction; and
- verification of uterine tone and, if the uterus is not well contracted, uterine massage.

The most important intervention to reduce postpartum hemorrhage is the immediate postpartum administration of a uterotonic within one minute of birth. Controlled cord traction is optional and should only be performed by a skilled provider.

IMMEDIATE POSTPARTUM UTEROTONIC

- Immediately after the birth of the baby, palpate the abdomen to rule out the presence of an additional baby(s).
- In the absence of an additional baby(s), **give oxytocin 10 units IM.**

Make sure there is no additional baby(s) before giving an injectable uterotonic medication IM or giving misoprostol orally.

Note: Oxytocin is preferred because it is effective two to three minutes after injection, has minimal side effects and can be used in all women.

- **If oxytocin is not available, give:**
 - oral misoprostol 600 mcg;
 - OR ergometrine (0.2 mg IM) or methylergometrine;
 - OR the fixed drug combination of oxytocin and ergometrine (1 mL = 5 IU oxytocin + 0.5 mg ergometrine).

Do not give ergometrine to women with pre-eclampsia, eclampsia or high blood pressure because it increases the risk of convulsions and cerebrovascular accidents.

CONTROLLED CORD TRACTION

The placenta may be allowed to deliver physiologically or can be delivered via controlled cord traction by a skilled provider. Controlled cord traction is contraindicated in settings without a skilled birth attendant.

- After cutting the cord, clamp the cord close to the perineum using sponge forceps. Hold the clamped cord and the end of the forceps with one hand.
- Place the other hand just above the woman's pubic bone and stabilize the uterus by applying counter-traction during controlled cord traction. This helps prevent inversion of the uterus.

Never apply cord traction (pull) without applying counter-traction (push) above the pubic bone with the other hand.

- Keep slight tension on the cord and await a strong uterine contraction (two to three minutes).
- When the **uterus becomes rounded (globular) or the cord lengthens**, very gently pull downward on the cord to deliver the placenta. Do not wait for a gush of blood before applying traction on the cord. Continue to apply counter-traction to the uterus with the other hand.
- If the **placenta does not descend** (i.e. there are no signs of placental separation) during 30 to 40 seconds of controlled cord traction, do not continue to pull on the cord. Instead:
 - Gently hold the cord and wait until the uterus is well contracted again. If necessary, use a sponge forceps to clamp the cord closer to the perineum as it lengthens.
 - With the next contraction, repeat controlled cord traction with counter-traction.
- As the placenta delivers, the thin membranes can tear off. Hold the placenta in two hands and gently turn it several times in a clockwise fashion until the membranes are twisted.
- Slowly pull to complete the delivery.

- If the **membranes tear**, gently examine the upper vagina and cervix while wearing sterile gloves. Use a sponge forceps to remove any pieces of membrane that are present.
- Examine the placenta carefully to be sure none of it is missing. If a **portion of the maternal surface is missing or if there are torn membranes with vessels**, suspect retained placental fragments (page S-44).
- If **uterine inversion occurs**, reposition the uterus (page P-109).
- If the **cord is pulled off**, manual removal of the placenta may be necessary (page P-91).
- Dispose of the placenta in a correct, safe and culturally appropriate manner.

VERIFICATION OF UTERINE TONE

- Immediately assess uterine tone. If **the uterus is soft**, massage the fundus of the uterus through the woman's abdomen until the uterus is contracted.

Note: Sustained uterine **massage** is not recommended as an intervention to prevent postpartum haemorrhage in women who have received a prophylactic uterotonic.

- Ensure that the uterus does not become relaxed (soft) after you stop uterine massage.
- Teach the woman how to assess uterine tone and massage her own uterus should it become soft.
- Estimate and record blood loss.

EXAMINATION FOR TEARS

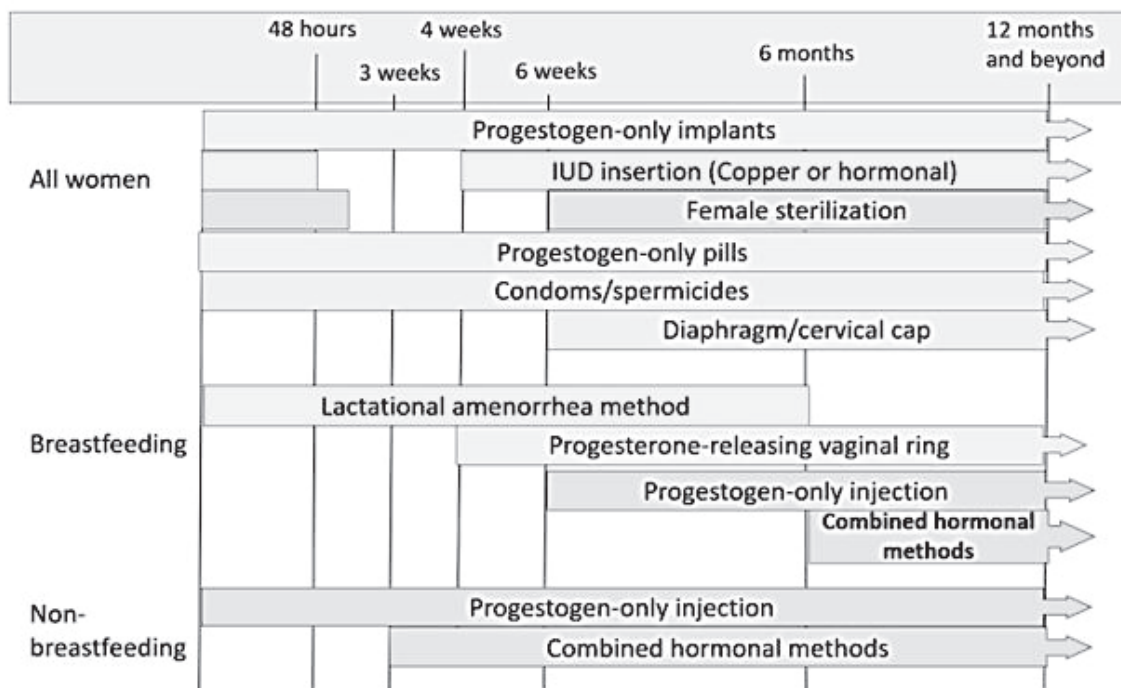
- Examine the woman carefully and repair any tears to the cervix (**page P-95**) or vagina (**page P-97**), or repair the episiotomy (**page P-87**).

POSTPARTUM FAMILY PANNING (PPFP)

- If a woman is medically eligible, has been appropriately counselled and has chosen postpartum insertion of an intrauterine device (IUD), either copper or levonorgestrel, proceed to post-placental insertion of the IUD just after expulsion and verification of an intact placenta. Within 10 mins of delivery till 48 hours postpartum (before discharge home, or after 1 month
- Subcutaneous Levenorgestrel or Etonogestrel Implants can also be inserted immediately after delivery if the woman has chosen this method of Family planning

Summary Contraceptive method eligibility postpartum

MEC 2015



CARE OF THE WOMAN AND NEWBORN DURING THE FIRST TWO HOURS AFTER CHILDBIRTH/BIRTH

- Keep the woman and her newborn in the birthing room for at least one hour after delivery of the placenta.

Avoid separating the woman from her baby whenever possible. Do not leave them unattended at any time.

- Conduct an examination of the woman and newborn before transferring them to the postpartum ward.
- Ensure that the room is warm (25°C).
- **Do not** discharge the woman and her newborn from the facility earlier than 24 hours after the birth.

CARE OF THE WOMAN

- Clean the woman and the area beneath her. Put a sanitary pad or folded clean cloth under her buttocks to collect blood. Help her change clothes if necessary.
- Monitor the woman's condition every 15 minutes for the first two hours and respond immediately if there are abnormal findings:
 - Assess uterine **tone** for early identification of uterine atony. Massage the uterus if atonic. Ensure that the uterus does not become relaxed (soft) after you stop uterine massage.
 - Measure blood pressure and pulse.
 - Measure and record blood loss.

Note: If bleeding is excessive, investigate the cause(s) for the bleeding (uterine atony, vaginal/cervical laceration, retained placenta) and treat appropriately (see **Vaginal bleeding after childbirth, page S-29**).

- For emergency signs, assess the woman using rapid assessment and manage per protocols.

- Encourage the woman to empty her bladder and ensure that she has passed urine.
- Encourage the woman to eat, drink and rest.
- Advise the woman on postpartum care, nutrition and hygiene.
- Ask the woman's companion to watch her and call for help if bleeding or pain increases, or if the woman feels dizzy or has severe headaches, visual disturbance or epigastric distress.
- Check the woman's record and give any treatment or prophylaxis that is due.

Do **not** give routine prophylactic antibiotic treatment in the absence of clinical signs of infection or recommended treatment criteria.

Routine antibiotic prophylaxis is **not** recommended for:

- women with uncomplicated vaginal birth
- women undergoing operative vaginal birth
- women with episiotomy
- women with first or second degree lacerations.

Routine antibiotic prophylaxis **is** recommended for:

- repair of third and fourth degree tears
- premature prelabour rupture of membranes
- caesarean birth
- manual removal of placenta or placement of uterine balloon tamponade.

- Counsel the woman on healthy timing and spacing of pregnancy, the lactational amenorrhoea method, return to fertility, and resumption of sexual relations.
- Counsel the woman on family planning methods that she can initiate immediately after birth:
 - lactational amenorrhoea method;
 - copper or levonorgestrel IUD (up to 48 hours after childbirth and then four weeks or more after childbirth); tubal ligation (up to one week after childbirth and then six weeks or more after childbirth);
 - progestin-only implants can be inserted immediately after childbirth
 - progestin-only oral contraceptive pills;
 - condoms;
 - vasectomy;
 - DepoProvera (only for women who are not breastfeeding).

CARE OF THE NEWBORN NOT REQUIRING RESUSCITATION

- When providing newborn care, ensure strict adherence to infection prevention and control principles (page **C-113**).
- Place the baby naked between the mother's breasts. Cover with a clean, dry drape or cloth, and cover the baby's head. Skin-to-skin contact should continue for at least one hour.
 - Only interrupt skin-to-skin contact for essential care.
 - Prevent hypothermia in the newborn when the baby is not in skin-to-skin contact with the mother.
- All newborns, including low birth weight babies who are able to breastfeed, should be put to the breast as soon as possible after birth when they are clinically stable and the mother and baby are ready.
- Encourage the mother to initiate breastfeeding within one hour after birth:
 - Encourage early and exclusive breastfeeding when the baby appears ready (begins "rooting").
 - To encourage early breastfeeding, position the baby near the mother's breasts, where the baby can latch when ready to feed.
 - Do not force the baby to the breast.
- Monitor the baby's condition every 15 minutes for the first two hours after birth and respond immediately if there are abnormal findings:
 - Check the baby's breathing and colour:

If the **baby becomes cyanotic** (bluish) or is **having difficulty breathing** (less than 30 or more than 60 breaths per minute), give oxygen by nasal catheter or prongs (**page S-173**).
 - Estimate the baby's body temperature by feeling the baby's skin (foot or forehead):
 - If the **baby's feet feel cold**, check the baby's axillary temperature;
 - If the **baby's temperature is less than 36.5°C**, rewarm the baby (**page S-178**).
 - Check the cord for bleeding.

- If the **cord is bleeding**, retie the cord more tightly.
 - For emergency signs, assess the baby using rapid assessment and respond immediately (**page S-165**).
- Wipe off any meconium or blood from the baby's skin. **Do not** remove vernix.
- Bathing should be delayed until 24 hours after birth (and should not be given less than six hours after birth).
- Assess breastfeeding:
 - Let the infant breastfeed on demand, if able to suck.
 - If **problems with breastfeeding are present**, observe breastfeeding, looking for signs of poor attachment.
 - Teach mothers how to recognize and manage common problems that may prevent successful breastfeeding.
- Ask the mother/parents to watch the baby and call for help if **the baby is having breathing difficulties, the baby's temperature is too low or high, there is bleeding from the cord, or there are convulsions**.
- Provide the following care:
 - Apply antimicrobial drops (1% silver nitrate solution or 2.5% povidone-iodine solution) or ointment (1% tetracycline ointment) to both of the baby's eyes once, according to national guidelines.

Note: Povidone-iodine should not be confused with tincture of iodine, which could cause blindness if used.

- Provide cord care. Keep the umbilical cord clean and dry. Nothing should be placed on the cord unless recommended by the health authority (for example, chlorhexidine).
- Weigh the baby and initiate special care for babies weighing less than 2500 g (**page C-112**).

- Give vitamin K by IM injection (1 mg/0.5 mL for term babies; 0.4 mg/kg for a maximum dose of 1 mg for preterm babies).
- Check the mother's record and give antibiotic prophylaxis to infants who have been exposed to risk factors for infection (i.e. preterm prelabour rupture of membranes; membranes ruptured more than 18 hours before birth; mother had fever higher than 38°C before birth or during labour; amniotic fluid was foul smelling or purulent; or mother has documented colonization with Group B streptococcus) (**page S-182**).
- Check the mother's record and give any prophylaxis that is due if the newborn has been exposed to syphilis (**page S-183**), tuberculosis (**page S-184**) or HIV (**page S-185**).
- Give oral polio, hepatitis B, BCG and other vaccinations, depending on national guidelines.

SAFE CHILDBIRTH CHECKLIST

Of the more than 130 million births occurring each year, an estimated 303 000 result in the mother's death, 2.6 million in stillbirth and another 2.7 million in a newborn death within the first 28 days of birth.

In response to unacceptably high rates of maternal and perinatal deaths occurring during labour and childbirth, the WHO developed the *WHO Safe Childbirth Checklist* (WHO, 2015) to support the delivery of essential maternal and perinatal care practices. The checklist addresses the major causes of maternal death (haemorrhage, infection, obstructed labour and hypertensive disorders), intrapartum-related stillbirths (inadequate intrapartum care) and neonatal deaths (birth asphyxia, infection and complications related to prematurity). It was developed following a rigorous methodology and tested for usability in 10 countries across Africa and Asia.

The *WHO Safe Childbirth Checklist Implementation Guide* (WHO, 2015) for health facilities has been developed to help birth attendants and health care leaders successfully launch and sustain use of the safe childbirth checklist.

See Appendix *WHO Safe Childbirth Checklist*

When a baby is born to a mother being treated for complications, management of the newborn is based on:

- whether the baby has a condition or problem requiring rapid treatment;
- whether the mother's condition permits her to care for her newborn completely, partially or not at all; and
- whether the mother's complication might have an impact on the newborn's health.

NEWBORN BABIES WITH URGENT PROBLEMS

- If the **newborn has an acute problem that requires treatment**, all health care providers who might come in contact with the newborn in the first 24 hours of life must recognize and give at least initial care for the problem (**page S-165**). Problems or conditions of the newborn requiring urgent interventions include:
 - gasping or not breathing;
 - breathing with difficulty (less than 30 or more than 60 breaths per minute, severe indrawing of the chest, or grunting);
 - central cyanosis (blueness);
 - pallor;
 - oozing or bleeding from cord;
 - drowsiness or unconsciousness;
 - movement only when stimulated or no movement at all;
 - not feeding well, inability to suck;
 - hypothermia (axillary temperature less than 36.5°C);
 - hyperthermia (axillary temperature greater than 37.5°C);
 - convulsions; and
 - severe jaundice (appears on the face during the first day of life or extends to the palms and soles at any time).

NEWBORN BABIES REQUIRING ADDITIONAL CARE

- The following babies require special care:
 - preterm babies (less than 37 complete weeks' gestation);
 - very low birth weight babies (less than 1500 g);
 - low birth weight babies (1500–2500 g); and
 - babies with obvious birth defects (e.g. cleft palate, ambiguous genitalia, spina bifida).
- If the **newborn has a malformation or other problem that does not require urgent care**:
 - Provide routine initial newborn care (**page C-107**).
 - Transfer the baby to the appropriate service to care for sick newborns as quickly as possible (**page C-114**).
- Asymptomatic (without symptoms) babies with the following circumstances are classified as at risk of infection and require early treatment (**page S-182**):
 - membranes ruptured more than 18 hours before birth;
 - mother is being treated with antibiotics for infection;
 - mother has fever greater than 38°C;
 - mother has confirmed maternal colonization with Group B streptococcus without adequate antibiotic therapy during labour;
 - mother is infected with HIV (**page S-185**) and/or syphilis (**page S-183**) and/or hepatitis B;
 - mother started tuberculosis treatment less than two months before birth (**page S-184**).

Infants exposed to infection should be referred to the appropriate service for the care of newborns.

NEWBORN BABIES WITHOUT PROBLEMS

- If the **newborn has no apparent problems**, provide routine initial newborn care and support early and exclusive breastfeeding (**page C-107**).

- If the **mother's condition permits**, keep the baby in skin-to-skin contact with the mother for at least the first hour after birth.

NOTE: Babies with birth weight less than 2000 g who are clinically stable should be given kangaroo mother care early in the first week of life.

- If the **mother's condition does not permit her to maintain skin-to-skin contact** with the baby after childbirth:
 - Wrap the baby in a soft, dry cloth, cover with a blanket and ensure that the baby's head is covered to prevent heat loss;
 - Observe the baby at least every 15 minutes for the first two hours after birth.
- If the **mother's condition requires prolonged separation** from the baby, transfer the baby to the appropriate service for the care of newborns (page C-114).

STANDARD INFECTION PREVENTION AND CONTROL PRACTICES WHEN CARING FOR A NEWBORN

- Avoid unnecessary separation of the newborn from the mother.
- Observe strict procedures for handwashing or alcohol handrubs for all staff and for families before and after handling infants.
- Provide appropriate umbilical cord care.
- Provide appropriate eye care.
- Promote exclusive breastfeeding.
- Use kangaroo mother care and avoid the use of incubators for preterm infants. If an incubator is used, do not use water (where *Pseudomonas* will easily colonize) for humidification, and ensure that the incubator is thoroughly cleaned with an antiseptic.
- Observe strict sterility for all procedures.
- Observe clean injection practices.

TRANSFERRING NEWBORN BABIES

- Explain to the mother and/or her companion why the baby is being transferred (**page C-5**).
- Keep the baby warm. Wrap the baby in a soft, dry cloth; cover with a blanket and ensure that the baby's head is covered to prevent heat loss. Babies with birth weight less than 2000 g should be transferred in skin-to-skin contact with the mother.
- Transfer the baby with the mother and a health care provider, if possible. If the **baby requires special treatment such as oxygen**, transfer the baby in an incubator or bassinet.
- Initiate breastfeeding as soon as the baby is ready to suckle or as soon as the mother's condition permits.
- If **breastfeeding has to be delayed** due to maternal or newborn problems, teach the mother to express breastmilk as soon as possible and ensure that this milk is given to the newborn with a spoon.
- Ensure that the service caring for the newborn receives the records of labour and childbirth and of any treatments given to the mother and newborn.

CREATING AN IMPROVED HEALTH CARE ENVIRONMENT

The district hospital should strive to create a welcoming environment for women, community members and health care providers from peripheral health units. It should support the efforts of other providers and work with them to correct deficiencies.

When dealing with other health care providers, doctors and midwives at the district hospital should:

- encourage and thank providers who refer patients, especially in the presence of women and their families;
- offer clinical guidance and corrective suggestions in private, so as to maintain the provider's credibility in the community;
- involve the provider (to an appropriate extent) in the continued care of the woman after discharge.

When dealing with community members, doctors and midwives at the district hospital should:

- promote patient, family and community involvement in strategic planning and improvement activities;
- invite members of the community to be part of the district hospital or health development committee;
- identify key persons in the community and invite them to the facility to learn about its function, as well as its constraints and limitations;
- create opportunities for the community to view the district hospital as a wellness facility (e.g. through vaccination campaigns and screening programmes).

MEETING THE NEEDS OF WOMEN

To enhance its appeal to women and the community, the district hospital should examine its own service delivery practices. The facility should create a culturally sensitive and comfortable environment that:

- respects women's modesty and privacy;
- welcomes family members; and
- provides a comfortable place for women and/or newborns (e.g. birthing bed in lowest possible position; warm and clean room).

With careful planning, the facility can create this environment without interfering with its ability to respond to complications or emergencies

IMPROVING REFERRAL PATTERNS

Health systems need to have formal tools and systems to facilitate referral and counter-referral and ensure continuity of care. Each woman who is referred to the district hospital should be given a standard referral slip containing the following information:

- general patient information (name, age, address);
- obstetrical history (parity, gestational age, complications in the antenatal period);
- relevant past obstetrical complications (e.g. previous caesarean, postpartum haemorrhage);
- relevant medical and surgical history;
- the specific problem for which she was referred; and
- treatments applied thus far and the results of those treatments.

Include the outcome of the referral on the referral slip. **Send the referral slip back to the referring facility** (counter-referral) with the woman or the person who brought her. Both the district hospital and the referring facility should keep a record of all referrals as a quality assurance mechanism:

- Referring facilities can assess the success and appropriateness of their referrals;
- The district hospital can review the records for patterns indicating that a health care provider or facility needs additional technical support or training.

PROVIDING TRAINING AND SUPPORTIVE SUPERVISION

District hospitals should offer high-quality, **participatory** clinical training for peripheral providers. Participatory training is skill-focused and more effective than classroom-based training because it:

- improves the relationship between health care providers at the district hospital and auxiliary and multipurpose workers from peripheral units;
- increases the familiarity of the peripheral providers with the clinical care provided at the district hospital; and

- promotes team building and facilitates supervision of health workers once they return to their community to implement the skills they have learned.

Shock is characterized by failure of the circulatory system to maintain adequate perfusion of the vital organs. Shock is a **life-threatening condition** that requires **immediate and intensive treatment**.

Suspect or anticipate shock if one or more of the following is present:

- bleeding in early pregnancy (e.g. abortion, ectopic or molar pregnancy);
- bleeding in late pregnancy or labour (e.g. placenta praevia, abruptio placentae, ruptured uterus);
- bleeding after childbirth (e.g. ruptured uterus, uterine atony, tears of the genital tract, retained placenta or placental fragments);
- infection (e.g. unsafe or septic abortion, amnionitis, endometritis, acute pyelonephritis);
- trauma (e.g. injury to uterus or bowel during abortion, ruptured uterus, tears of genital tract).

SYMPTOMS AND SIGNS f

Diagnose shock if the following symptoms and signs are present:

- fast, weak pulse (110 beats per minute or more);
- low blood pressure (systolic less than 90 mmHg).

Other symptoms and signs of shock include:

- pallor (especially of inner eyelid, palms or around mouth);
- sweatiness or cold, clammy skin;
- rapid breathing (rate of 30 breaths per minute or more);
- anxiousness, confusion or unconsciousness;
- scanty urine output (less than 30 mL per hour).

IMMEDIATE MANAGEMENT

When managing the woman's problem, apply basic principles when providing care (page C-25).

- **SHOUT FOR HELP.** Urgently mobilize all available personnel.
- Monitor vital signs (pulse, blood pressure, respiration, temperature).
- If the **woman is unconscious**, turn her onto her side to minimize the risk of aspiration if she vomits, and to ensure that an airway is open.
- Keep the woman warm but do not overheat her, as this will increase peripheral circulation and reduce blood supply to the vital centres.
- Elevate the legs to increase return of blood to the heart (if possible, raise the foot end of the bed).

SPECIFIC MANAGEMENT

- Start an IV infusion (two if possible) using a large-bore (16-gauge or largest available) cannula or needle.
- Collect blood for estimation of haemoglobin, immediate cross-match and bedside clotting test (**page S-3**), just before infusion of fluids:
 - Rapidly infuse IV fluids (normal saline or Ringer's lactate) initially at the rate of 1 L in 15–20 minutes.

Note: Avoid using plasma substitutes (e.g. dextran). There is no evidence that plasma substitutes are superior to normal saline in the resuscitation of a shocked woman, and dextran can be harmful in large doses.
 - Give at least 2 L of these fluids in the first hour. This is over and above fluid replacement for ongoing losses.

Note: A more rapid rate of infusion is required in the management of shock resulting from bleeding. Aim to replace two to three times the estimated fluid loss.

Do not give fluids by mouth to a woman in shock.

- If a **peripheral vein cannot be cannulated**, perform a venous cutdown (**Fig. S-1. Page S-4**).
- Continue to monitor vital signs (every 15 minutes) and blood loss.
- Catheterize the bladder and monitor fluid intake and urine output.

- Give oxygen at 6–8 L per minute by mask or nasal cannulae.

If available, apply a non-pneumatic anti-shock garment (NASG) as a temporizing measure until appropriate care is available (**page S-39**).

BEDSIDE CLOTTING TEST

Assess clotting status using this **bedside clotting test**:

- Take 2 mL of venous blood into a small, dry, clean, plain glass test tube (approximately 10 mm x 75 mm).
- Hold the tube in a closed fist to keep it warm ($\pm 37^{\circ}$).
- After four minutes, tip the tube slowly to see if a clot is forming. Then tip it again every minute until the blood clots and the tube can be turned upside down.
- Failure of a clot to form after seven minutes or a soft clot that breaks
- down easily suggests coagulopathy (**page S-24**).

Early Recognition of Shock

Early recognition of Shock (reduced tissue perfusion) in PPH
Changes in vital signs appear late (>1000 mls bld loss)

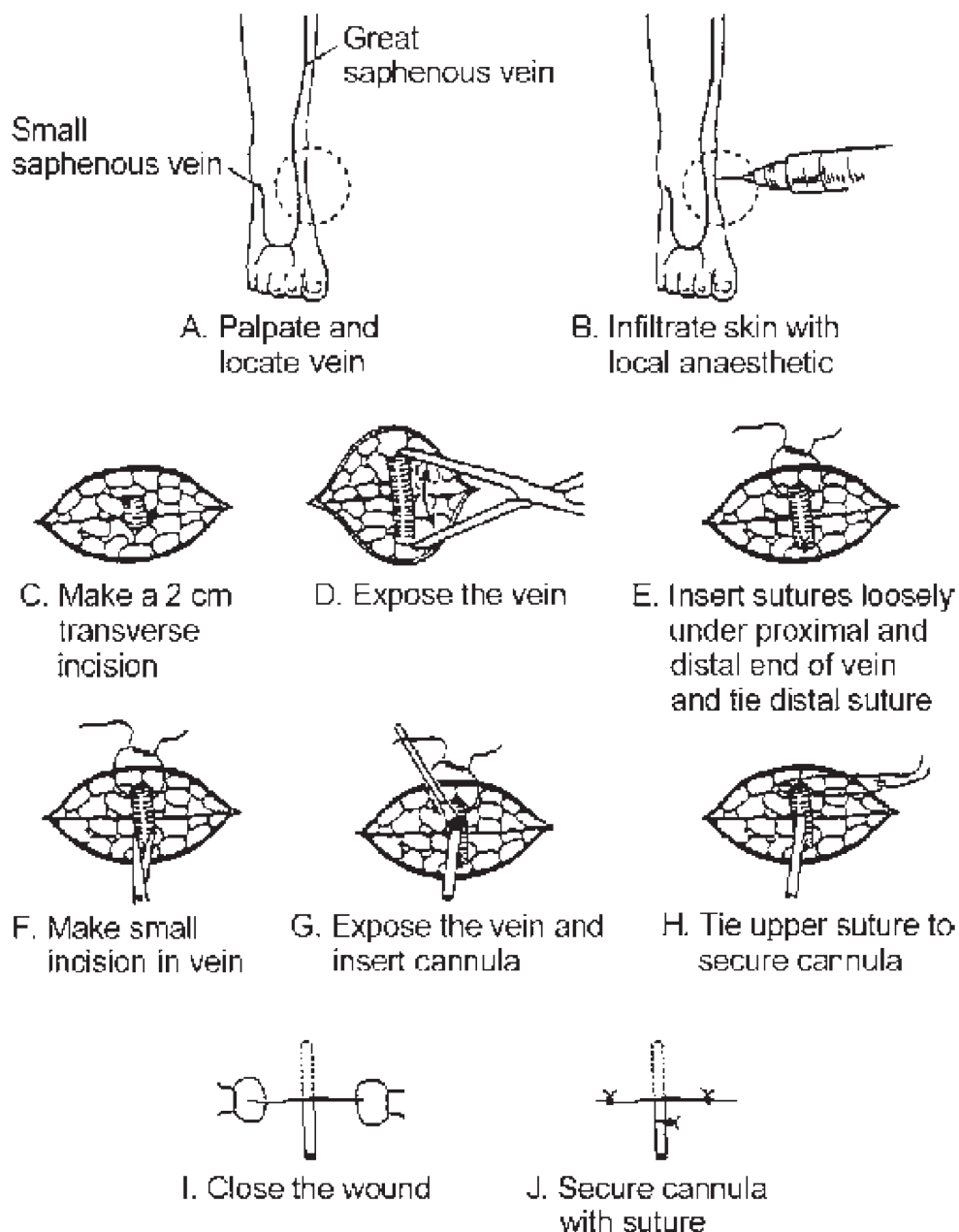
Shock Index (SI)

- Heart Rate: Systolic BP
- 0.7 – 0.9 (standard Obstetric)
- 0.5 - 0.7 (non pregnant)
- >0.9 = haemodynamic instability
 - Increased mortality
- >1 = Blood Tx increased risk

Rule of 30

- Loss of 30% blood vol
 - 70mls/kg adults
 - 100 mls/kg pregnant women
- 30% fall in Hematocrit
- 30% fall in Hb (3 g/dl)
- 30 mmHg fall in Systolic BP
- 30 beats/min rise in pulse

FIGO recommends use of Shock Index in Diagnosis & Management of PPH

FIGURE S-1. Venous cutdown

DETERMINING AND MANAGING THE CAUSE OF SHOCK

Determine the cause of shock after the woman is stabilized.

- If **heavy bleeding is suspected** as the cause of shock:
 - Take steps simultaneously to stop bleeding (e.g. uterotonic drugs, uterine massage, bimanual compression, uterine balloon tamponade, aortic compression, preparations for surgical intervention).

- Transfuse as soon as possible to replace blood loss (**page C-37**).
 - Determine the cause of bleeding and manage accordingly:
 - If **bleeding occurs during first 22 weeks of pregnancy**, suspect abortion or ectopic or molar pregnancy (**page S-17**);
 - If **bleeding occurs after 22 weeks or during labour but before childbirth**, suspect placenta praevia (**page S-25**), abruptio placentae (**page S-23**) or ruptured uterus (**page S-24**);
 - If **bleeding occurs after childbirth**, suspect ruptured uterus, uterine atony, tears of genital tract, retained placenta or placental fragments (**page S-29**).
 - Reassess the woman's condition for signs of improvement (**page S-6**).
 - If **infection is suspected** as the cause of shock:
 - Collect appropriate samples (blood, urine, pus) for microbial culture, if facilities are available, before starting antibiotics.
 - Give the woman a combination of antibiotics to cover aerobic and anaerobic infections and continue until she is fever-free for 48 hours (**page C-50**):
 - ampicillin 2 g IV every six hours;
 - PLUS gentamicin 5 mg/kg body weight IV every 24 hours.
- Do not give antibiotics by mouth to a woman in shock.**
- Reassess the woman's condition for signs of improvement (**page S-5**).
 - If **trauma is suspected** as the cause of shock, prepare for surgical intervention.

REASSESSMENT

- Reassess the woman's response to fluids within 30 minutes to determine if her condition is improving. Signs of improvement include:
 - stabilizing pulse (rate of 90 per minute or less);

- increasing blood pressure (systolic 100 mmHg or more);
 - improving mental status (less confusion or anxiety);
 - increasing urine output (30 mL per hour or more).
- If the **woman's condition improves**:
 - Adjust the rate of infusion of IV fluids to 1 L in six hours.
 - Continue management for the underlying cause of shock (**page S-4**).
- If the **woman's condition fails to improve or stabilize**, provide further management (see below).

FURTHER MANAGEMENT

- Continue to infuse IV fluids, adjusting the rate of infusion to 1 L in six hours, and maintain oxygen at 6–8 L per minute.
- Closely monitor the woman's condition.
- Perform laboratory tests, including repeat haemoglobin determination, blood grouping and Rh typing. **If facilities are available**, check serum electrolytes, serum creatinine and blood pH.

PROBLEM

- Vaginal bleeding occurs during the first 22 weeks of pregnancy.

IMMEDIATE MANAGEMENT

When managing the woman's problem, apply basic principles when providing care (page C-25).

- Perform a **rapid evaluation** of the woman's general condition, vital signs (pulse, blood pressure, respiration), level of consciousness, presence of anxiety and/or confusion, blood loss, and colour and temperature of skin (**page C-1**).
- If **shock is suspected**, immediately begin treatment (**page S-1**). Even if signs of shock are not present, keep shock in mind as you evaluate the woman further because her status may worsen rapidly. If **shock develops**, it is important to begin treatment immediately.

If the **woman is in shock**, consider ruptured ectopic pregnancy (**Table S-7, page S-16**).

- Depending on gestational age, check the fetal heart rate and ask about fetal movements:
 - If there are **fetal heart rate abnormalities** (less than 100 or more than 180 beats per minute), suspect fetal distress (**page S-109**).
 - If **fetal heart cannot be heard**, ask several other persons to listen or use a Doppler stethoscope, if available. If **fetal heart cannot be heard, suspect fetal death** (**page S-156**).
- Start an IV infusion and infuse IV fluids (**page C-34**).

Note: Send a blood sample for haemoglobin or haematocrit and type and screen before infusing IV fluids.

DIAGNOSIS

- **Consider ectopic pregnancy** in any woman with anaemia, pelvic inflammatory disease, threatened abortion or unusual complaints about abdominal pain.

Note: If **ectopic pregnancy is suspected**, perform bimanual examination gently because an early ectopic pregnancy is easily ruptured.

- When available, perform ultrasound with quantitative assessment of beta-human chorionic gonadotropin as the essential diagnostic test.
- **Consider abortion** in any woman of reproductive age who has a missed period (delayed menstrual bleeding with more than one month having passed since her last menstrual period) and has one or more of the following: bleeding, cramping, partial expulsion of products of conception, dilated cervix or smaller uterus than expected.
- If **abortion is a possible diagnosis**, identify and treat any complications immediately (**Table S-2, page S-9**).

TABLE S-1. Diagnosis of vaginal bleeding in early pregnancy

Presenting Symptom and Other Symptoms and Signs Typically Present	Symptoms and Signs Sometimes Present	Probable Diagnosis
• Light bleeding^a • Closed cervix • Uterus corresponds to dates	• Cramping/lower abdominal pain • Uterus softer than normal	Threatened abortion, page S-11
• Light bleeding^a • Abdominal pain • Closed cervix • Uterus slightly larger than normal • Uterus softer than normal	• Fainting • Tender adnexal mass • Amenorrhoea • Cervical motion tenderness	Ectopic pregnancy, page S-15
• Light bleeding^a • Closed cervix • Uterus smaller than dates • Uterus softer than normal	• Light cramping/lower abdominal pain • History of expulsion of products of conception	Complete abortion, page S-13

Presenting Symptom and Other Symptoms and Signs Typically Present	Symptoms and Signs Sometimes Present	Probable Diagnosis
• Heavy bleeding^b • Dilated cervix • Uterus corresponds to dates	• Cramping/lower abdominal pain • Tender uterus • No expulsion of products of conception	Inevitable abortion, page S-12
• Heavy bleeding^b • Dilated cervix • Uterus smaller than dates	• Cramping/lower abdominal pain • Partial expulsion of products of conception	Incomplete abortion, page S-12
• Heavy bleeding^b • Dilated cervix • Uterus larger than dates • Uterus softer than normal • Partial expulsion of products of conception, which resemble grapes	• Nausea/vomiting • Spontaneous abortion • Cramping/lower abdominal pain • Ovarian cysts (easily ruptured) • Early onset pre-eclampsia • No evidence of a fetus	Molar pregnancy, page S-17

^a Light bleeding: takes five minutes or longer for a clean pad or cloth to be soaked

^b Heavy bleeding: takes less than five minutes for a clean pad or cloth to be soaked and/or large blood clots are expelled.

TABLE S-2. Diagnosis and management of complications of abortion

Symptoms and Signs	Complication	Management
• Lower abdominal pain • Rebound tenderness • Tender uterus • Prolonged bleeding • Malaise • Fever • Foul-smelling vaginal discharge • Purulent cervical discharge • Cervical motion tenderness	Infection/sepsis	Begin antibiotics ^a as soon as possible before attempting manual vacuum aspiration (page P-75).

Symptoms and Signs	Complication	Management
• Cramping/abdominal pain • Rebound tenderness • Abdominal distension • Rigid (tense and hard) abdomen • Shoulder pain • Nausea/vomiting • Fever	Uterine, vaginal or bowel injuries	Perform a laparotomy to repair the injury and perform manual vacuum aspiration (page P-75) simultaneously. Seek further assistance if required.

^a Clindamycin 600 mg IV every six to eight hours and gentamicin 5 mg/kg body weight IV every 24 hours. If clindamycin is not available, give ampicillin 2 g IV every six hours and gentamicin 5 mg/kg body weight IV every 24 hours.

BOX S-1. Types of abortion

Spontaneous abortion is defined as the loss of a pregnancy before fetal viability (22 weeks of gestation). The stages of spontaneous abortion may include:

- threatened abortion (pregnancy may continue);
- inevitable abortion (pregnancy will not continue and will proceed to incomplete/complete abortion);
- incomplete abortion (products of conception are partially expelled);
- complete abortion (products of conception are completely expelled).

Induced abortion is defined as a process by which pregnancy is terminated before fetal viability.

Unsafe abortion is defined as a procedure performed either by persons lacking necessary skills or in an environment lacking minimal medical standards, or both.

Septic abortion is defined as abortion complicated by infection. Sepsis may result from infection if organisms rise from the lower genital tract following either spontaneous or unsafe abortion. Sepsis is more likely to occur if there are retained products of conception and evacuation has been delayed. Sepsis is a frequent complication of unsafe abortion involving instrumentation.

SPECIFIC MANAGEMENT

If unsafe abortion is suspected, examine for signs of infection or uterine, vaginal or bowel injury (Table S-2, page S-9), and thoroughly irrigate the vagina to remove any herbs, local medications or caustic substances.

THREATENED ABORTION

- Medical treatment is usually not necessary.
- Advise the woman to avoid strenuous activity and sexual intercourse, but bed rest is not necessary.
- If **bleeding stops**, follow up in an antenatal clinic. Reassess if bleeding recurs.
- If **bleeding persists**, assess for fetal viability (pregnancy test/ultrasound) or ectopic pregnancy (ultrasound). Persistent bleeding, particularly in the presence of a uterus that is larger than expected, may indicate twins or molar pregnancy.

TABLE S-3. Management options for threatened abortion by gestational age

Surgical Management		Medical Management		Expectant Management
Less than 12–14 weeks	More than 12–14 weeks	Less than 12–14 weeks	More than 12–14 weeks	Any gestational age less than 24 weeks
Not applicable (NA)	NA	NA	NA	Medical treatment usually not necessary (page S-12)

Do not give medications such as hormones (e.g. estrogens or progestins) or tocolytic agents (e.g. salbutamol or indomethacin), as they will not prevent abortion.

INEVITABLE AND INCOMPLETE ABORTION

BOX S-2. Managing inevitable and incomplete abortion

- In cases of inevitable and incomplete abortion, surgical evacuation and medical or expectant management are all reasonable management options. To determine the best course of action, one must take the following into consideration:
 - the urgency of the particular case (i.e. haemodynamic stability);
 - the skill set of the available providers and staff;
 - the health facility setting and available materials, supplies and medications; and
 - the woman's preference, after complete counselling on options.
- **Evacuation: If evacuation is appropriate and the woman prefers surgical management**, examine the woman to determine if products of conception are visible in the os or the vagina and remove them with sponge or ring forceps. **If products are not visible**, proceed with manual vacuum aspiration if gestational age is less than 12–14 weeks (**page P-75**). Proceed with dilatation and evacuation (D and E) (**page S-18**) if gestational age is more than 12–14 weeks and the majority of the products remain in the uterus.
- **Medical Management:** If medical management is determined to be appropriate (i.e. the woman is haemodynamically stable and ectopic pregnancy is not suspected), and it is the woman's preference, proceed with a medical protocol.
 - **See Table S-4 (page S-13)** for medical management protocols. Optimal doses for misoprostol in the setting of incomplete abortion have not been determined. However, a reasonable protocol can be found in the management table.
 - **If misoprostol is not available**, oxytocin is an option for medical management after 16 weeks.
 - **If expulsion of products of conception is not achieved**, proceed to surgical evacuation.
 - Ensure follow-up of the woman after treatment (**page S-14**).
- **Expectant Management:** Expectant management is a reasonable option if the woman is haemodynamically stable and prefers to avoid surgical or medical intervention.
 - Await spontaneous expulsion of products of conception.
 - **If spontaneous expulsion does not occur in a reasonable timeframe**, depending on the woman's tolerance for the discomfort and wait and her haemodynamic status, proceed to either medical management or evacuation of the uterus via manual vacuum aspiration (**page P-75**) or D and E (**page S-18**).

TABLE S-4. Management options for inevitable and incomplete abortion by gestational age

Surgical Management		Medical Management		Expectant Management
Less than 12–14 weeks	More than 12–14 weeks	Less than 12–14 weeks	More than 12–14 weeks	Any gestational age less than 24 weeks
Inevitable abortion				
Manual vacuum aspiration (page P-75)	D and E (page S-18)	Misoprostol 800 mcg by vagina or sublingual every 3–12 hours; maximum three doses	Misoprostol 400 mcg by vagina or sublingual every three hours, maximum five doses OR Oxytocin 40 units in 1 L IV fluids at 40 drops per minute	Reasonable if woman is haemodynamically stable
Incomplete abortion				
Manual vacuum aspiration (page P-75)	D and E (page S-18)	Misoprostol 400 mcg sublingual or 600 mcg by mouth for one dose	Misoprostol 200 mcg by vagina every four hours until expulsion, maximum 800 mcg OR Oxytocin 40 units in 1 L IV fluids at 40 drops per minute	Reasonable if woman is haemodynamically stable

COMPLETE ABORTION

- Evacuation of the uterus is usually not necessary.
- Observe for heavy bleeding. If **heavy bleeding ensues**, proceed to manual vacuum aspiration to ensure that there are no remaining products, and administer 800 mcg of misoprostol for management of post-abortion haemorrhage.
- Ensure follow-up of the woman after treatment (see below).

TABLE S-5. Management options for complete abortion by gestational age

Surgical Management		Medical Management		Expectant Management
Less than 12–14 weeks	More than 12–14 weeks	Less than 12–14 weeks	More than 12–14 weeks	Any gestational age less than 24 weeks
Manual vacuum aspiration (page P-75) in case of heavy bleeding ^a	NA	NA	NA	NA

^a Heavy bleeding: takes less than five minutes for a clean pad or cloth to be soaked

OTHER KEY COMPONENTS OF CARE FOR WOMEN WHO HAVE HAD AN ABORTION

Before discharge, tell a woman who has had a spontaneous abortion that spontaneous abortion is common and occurs in at least 15% (one in every seven) of clinically recognized pregnancies. Reassure her that the chances for a subsequent successful pregnancy are good unless there has been sepsis or a cause of the abortion is identified that may have an adverse effect on future pregnancies (this is rare).

Some women may want to become pregnant soon after having an abortion. They should be encouraged to delay the next pregnancy until they have completely recovered.

Abortion care should always include comprehensive contraceptive counselling with initiation of the method of choice as soon as desired after the abortion.

Before the desired method is initiated, women should be assessed for medical eligibility (see **WHO, *Medical Eligibility Criteria for Contraceptive Use*, 5th ed., 2015**).

TABLE S-6. Family planning methods

Type of Contraceptive	Advise to Start
Hormonal (pills, ring, injections, implants)	• Immediately • No additional contraceptive protection is needed
Condom	• Immediately
Diaphragm, cap	• Unsuitable until six weeks after second-trimester abortion
Intrauterine device	• Immediately • If infection is present or suspected, delay insertion until cleared • If haemoglobin is less than 7 g/dL, delay until anaemia improves • Provide an interim method (e.g. condom)
Voluntary tubal ligation	• Immediately • If infection is present or suspected, delay surgery until cleared • If haemoglobin is less than 7 g/dL, delay until anaemia improves • Provide an interim method (e.g. condom)

Identify any other reproductive health services that a woman might need. For example, some women may need:

- a tetanus prophylaxis or tetanus booster;
- treatment for sexually transmitted infections; or
- cervical cancer screening.

ECTOPIC PREGNANCY

An ectopic pregnancy is one in which implantation occurs outside the uterine cavity. The fallopian tube is the most common site of ectopic implantation (greater than 90%).

Symptoms and signs are extremely variable, depending on whether or not the pregnancy has ruptured (**Table S-7, page S-16**). Culdocentesis (cul-de-sac puncture, **page P-81**) is an important tool for the diagnosis of ruptured ectopic pregnancy, but it is less useful than a serum pregnancy test combined with ultrasonography. If **nonclotting blood is obtained**, begin immediate management.

TABLE S-7. Symptoms and signs of unruptured and ruptured ectopic pregnancy

Unruptured Ectopic Pregnancy	Ruptured Ectopic Pregnancy
• Symptoms of early pregnancy (irregular spotting or bleeding, nausea, swelling of breasts, bluish discolouration of vagina and cervix, softening of cervix, slight uterine enlargement, increased urinary frequency) • Abdominal and pelvic pain	• Collapse and weakness • Fast, weak pulse (110 beats per minute or more) • Hypotension • Hypovolaemia • Acute abdominal and pelvic pain • Abdominal distension^a • Rebound tenderness • Pallor

^a A distended abdomen with shifting dullness may indicate free blood.

DIFFERENTIAL DIAGNOSIS

The most common differential diagnosis for ectopic pregnancy is threatened abortion. Others are acute or chronic pelvic inflammatory disease, ovarian cysts (torsion or rupture), and acute appendicitis.

If available, ultrasound can help to distinguish a threatened abortion or twisted ovarian cyst from an ectopic pregnancy.

IMMEDIATE MANAGEMENT

- Cross-match blood and arrange for immediate laparotomy. **Do not wait for blood before performing surgery.**
- At surgery, inspect both ovaries and fallopian tubes:
 - If there is **extensive damage to the tube**, perform salpingectomy (the bleeding tube and the products of conception are removed together). This is the treatment of choice in most cases (**page P-131**).
 - Rarely, if there is **little tubal damage**, perform salpingostomy (the products of conception can be removed and the tube conserved). Because the risk of another ectopic pregnancy is high, this should be done only when the conservation of fertility is very important to the woman (**page P-131**).

Autotransfusion

If **significant haemorrhage occurs**, autotransfusion can be used if the **blood is unquestionably fresh and free of infection** (in later stages of pregnancy, blood is contaminated [e.g. with amniotic fluid] and should not be used for autotransfusion). The blood can be collected before surgery or after the abdomen is opened:

- When the woman is on the operating table before surgery and the abdomen is distended with blood, it is sometimes possible to insert a needle through the abdominal wall and collect the blood in a donor set.
- Alternatively, open the abdomen:
 - Scoop the blood into a basin and strain through gauze to remove clots.
 - Clean the top portion of a blood donor bag with antiseptic solution and open it with a sterile blade.
 - Pour the woman's blood into the bag and reinfuse it through a filtered set in the usual way.
 - If a **donor bag with anticoagulant is not available**, add sodium citrate 10 mL to each 90 mL of blood.

SUBSEQUENT MANAGEMENT

- Before discharge, provide counselling and advice on prognosis for fertility. Given the increased risk of a future ectopic pregnancy, family planning counselling and provision of a family planning method, if desired, is especially important (**Table S-6, page S-15**).
- Correct anaemia with ferrous sulfate or ferrous fumarate 60 mg by mouth daily for three months.
- Schedule a follow-up visit at four weeks.

MOLAR PREGNANCY

Molar pregnancy is characterized by an abnormal proliferation of chorionic villi.

IMMEDIATE MANAGEMENT

- If the **diagnosis of molar pregnancy is certain**, cross-match blood if possible and arrange for evacuation of the uterus:

- **If cervical dilatation is needed**, use a paracervical block (**page P-1**).
- Use vacuum aspiration (**page P-75**). Manual vacuum aspiration is safer and associated with less blood loss than sharp metal curettage. The risk of perforation using the metal curette is high.
- Have three syringes prepared and ready for use during the evacuation. The uterine contents are copious and it is important to evacuate them rapidly.
- Infuse oxytocin 20 units in 1 L IV fluids (normal saline or Ringer's lactate) at a rate of 60 drops per minute to prevent haemorrhage once evacuation is under way.

SUBSEQUENT MANAGEMENT

- Recommend a hormonal family planning method for at least one year to prevent pregnancy (**Table S-6, page S-15**). Voluntary tubal ligation may be offered if the woman has completed her family.
- Follow up every eight weeks for at least one year with urine pregnancy tests because of the risk of persistent trophoblastic disease or choriocarcinoma. Assess the woman for irregular bleeding through a history and physical examination. If the **urine pregnancy test is not negative after eight weeks** or if it **becomes positive again** within the first year, urgently refer the woman to a tertiary care centre for further follow-up and management of choriocarcinoma.

DILATATION AND EVACUATION PROCEDURE

After 14 weeks, a manual vacuum aspiration procedure and cannula might not be sufficient to evacuate the remaining intrauterine contents. In this case, it is necessary to use a heavy forceps to complete the evacuation.

- Follow the manual vacuum aspiration procedure steps through placement of the tenaculum or vulsellum forceps.
- Leave the tenaculum or vulsellum forceps in place on the anterior lip of the cervix, or replace the forceps with a ring or sponge forceps if the tenaculum appears to be tearing the cervix.
- Use graduated dilators to further dilate the cervix if the forceps do not pass easily through the cervix (**page P-71**).
- Take care not to tear the cervix.